ENGINEERING / OPERATIONS

Payment Risk & Fraud Intelligence Platform: Engineering and Operations Documentation

System architecture, fraud model design, security, production runbook, incident reports and disaster recovery

Organization: Halcyon Payments, Inc.
Document set version: 2026.3 (Q3 2026 edition)
Status: CURRENT
Published: 2026-09-25
Owner: Marcus Lee, Head of Fraud Engineering
Classification: Internal - Confidential
Note: This is a synthetic document set created for testing an enterprise retrieval system. All names, figures and events are fictional.

Contents

Contents 2

Part 1. Document Control 5

1. About this document set 5

Part 2. System Architecture Design 8

1. Purpose, scope and principles 8
2. System overview 9
3. Component catalog 10
4. Event streaming and data architecture 19
5. Infrastructure and deployment 21
6. Performance and scalability 23
7. Failure scenarios and degradation 24
8. Architecture decisions 28
9. Version history 29

Part 3. Fraud Model Technical Design 30

1. FraudShield model family and status 30
2. Model design 32
3. Feature engineering 33
4. Real-time feature pipeline 42
5. Training and validation pipeline 42
6. Model serving and inference 43
7. Monitoring, thresholds and rollback conditions 43
8. Deployment process 44
9. Known limitations 45

Part 4. Security Architecture 46

1. Scope and security principles 46
2. Data classification 46
3. Identity and access management 47
4. Encryption and key management 48
5. Network security 48
6. Logging, audit and retention 48
7. Privacy 49

8. Security control matrix . . . 50
9. Access to documentation and the RAG assistant . . . 50

Part 5. Production Runbook . . . 51

1. Using this runbook . . . 51
2. Service catalog and objectives . . . 52
3. Alert catalog . . . 52
4. Standard procedures . . . 54
5. Diagnostic commands and queries . . . 59
6. Change windows and freezes . . . 59
7. Post-incident actions . . . 59

Part 6. Incident and Postmortem Reports . . . 60

1. Incident log . . . 60
2. INC-2026-041: August 2026 Fraud Platform Postmortem . . . 60
3. INC-2026-057: Feature pipeline failure . . . 64
4. Patterns across incidents . . . 67

Part 7. Disaster Recovery Architecture . . . 68

1. Strategy . . . 68
2. Replication and backup . . . 68
3. Failover test history . . . 69
4. Failure of the disaster recovery region . . . 69

Part 8. Technical Appendices . . . 70

1. Appendix A: Message schemas and configuration reference . . . 70
2. Appendix B: Observability and testing . . . 78
3. Appendix C: Model validation and evidence . . . 81
4. Appendix D: Additional runbook procedures . . . 84
5. Appendix E: Threat model and security assessments . . . 87
6. Appendix F: Earlier 2026 incident summaries . . . 88
7. Appendix G: Data Platform Architecture . . . 89
8. Appendix H: Production Support Guide . . . 92

Part 1. Document Control

1. About this document set

This engineering and operations documentation set describes the Payment Risk & Fraud Intelligence Platform (PRFIP) operated by Halcyon Payments, Inc. It combines five formal documents that were previously issued separately: the System Architecture Design, the Fraud Model Technical Design, the Security Architecture, the Production Runbook, and the Incident and Postmortem Reports. Each part carries its own owner, status and effective date, shown in the table below. Where two parts appear to disagree, the part with the later effective date and status APPROVED prevails, and the discrepancy should be reported to the document owner.

Part	Document	Version	Status	Effective date	Owner
2	System Architecture Design	v3.4	CURRENT (Approved)	2026-09-01	Marcus Lee
3	Fraud Model Technical Design (FraudShield v4.2)	v4.2.3	CURRENT (Approved)	2026-08-27	Sarah Johnson
4	Security Architecture	v2.6	CURRENT (Approved)	2026-06-15	Elena Petrova
5	Production Runbook	v5.1	CURRENT (Approved)	2026-09-10	Daniel Okafor
6	Incident and Postmortem Reports	Q3 2026 edition	CURRENT (Published)	2026-09-25	Daniel Okafor
7	Disaster Recovery Architecture	v2.2	CURRENT (Approved)	2026-07-20	Marcus Lee

Table 1.1. Parts of this document set with current status.

1.1 Superseded and draft versions

Earlier editions remain available in the document repository for audit purposes but must not be used for operational decisions. The following table lists the versions of the system architecture and model design documents that a reader may encounter.

Document	Version	Status	Effective	Notes
System Architecture Design	v3.4	CURRENT	2026-09-01	Adds schema compatibility enforcement and dead-letter routing after INC-2026-057
System Architecture Design	v3.3	ARCHIVED	2026-06-30	Superseded by v3.4; described 6 grouped feature lookups per request
System Architecture Design	v3.2	ARCHIVED	2026-02-10	Introduced 250 ms deadline (ADR-009)
System Architecture Design	v4.0	DRAFT	not effective	Proposes active-active multi-region (ADR-010); not approved
Fraud Model Technical Design	v3.1	ARCHIVED	2025-02-10	Describes FraudShield v3.1 (XGBoost, 61 features, thresholds 0.90 / 0.65)
Fraud Model Technical Design	v4.2.3	CURRENT	2026-08-27	Describes FraudShield v4.2 in production

Document	Version	Status	Effective	Notes
Fraud Model Technical Design	v5.0	DRAFT	not effective	Describes FraudShield v5.0 in shadow evaluation; thresholds are proposals only

Table 1.2. Versions of key documents and their status.

Rule of precedence: When asked for the current production configuration, use the document with status CURRENT and the latest effective date. A document with status DRAFT or PROPOSED describes intended future behavior and is not a source of truth for production. A document with status ARCHIVED describes a past configuration.

1.2 Roles and ownership

Name	Role	Responsibility
Marcus Lee	Head of Fraud Engineering	Platform architecture owner; chairs Architecture Review Board
Sarah Johnson	Lead Data Scientist, Fraud Models	Owner of FraudShield (all versions)
Priya Raman	Feature Platform Lead	Owner of Feature Service and Feature Store
Daniel Okafor	SRE Lead	Owner of Production Runbook; incident commander
Elena Petrova	Principal Security Architect	Owner of Security Architecture
Tom Alvarez	Fraud Operations Manager	Owner of manual review and case handling
Aisha Khan	Head of Model Risk & Validation	Independent validator; Model Governance Committee chair
Rahul Mehta	Data Governance Lead	Owner of data classification and retention
Grace Chen	Senior Product Manager	Owner of business requirements
Wei Zhang	Senior Platform Engineer	Owner of Kafka and Flink streaming layer
Naomi Fischer	Integration Lead, Payment Gateway	Owner of Payment Gateway integration
Ravi Menon	Site Reliability Engineer	Primary on-call engineer, August 2026 rotation

Table 1.3. People and areas of ownership referenced in this document set.

1.3 Glossary

Term	Meaning
PRFIP	Payment Risk & Fraud Intelligence Platform
FraudShield	The family of fraud scoring models; current production version is v4.2
v4.2-RF	Reduced-feature variant of FraudShield v4.2, trained on the 32 request-only features and used when the Feature Service is unavailable
Feature Service	Service that reads precomputed features from the online store at scoring time
Online store	Redis Cluster holding the latest feature values per entity
Degraded decision	A decision produced by the reduced-feature model or by rules only; marked degraded=true
PSI	Population stability index, a drift measure comparing two distributions
Shadow mode	A model version scores live traffic but its output is logged only and never returned to callers
Canary	Staged exposure of a new version to a small share of live traffic
MGC	Model Governance Committee, the approval authority for model versions and thresholds
ARB	Architecture Review Board, the approval authority for architecture decisions
RRR	Release Readiness Review
CHG	Production change request identifier
SEV	Incident severity from SEV-1 (critical) to SEV-4 (minor)
RTO / RPO	Recovery time objective and recovery point objective
DLQ	Dead-letter queue (topic) for messages that cannot be processed
PAN	Primary account number (full card number); never received or stored by PRFIP
Token	Substitute value standing in for a PAN, issued by the Token Vault
CDE	Cardholder data environment as defined by PCI DSS
3DS	3-D Secure cardholder authentication protocol
MCC	Merchant category code

Table 1.4. Glossary.

Part 2. System Architecture Design

Attribute	Value
Document	System Architecture Design
Version / status	v3.4 / CURRENT (Approved)
Effective date	2026-09-01
Owner	Marcus Lee, Head of Fraud Engineering
Approved by	Architecture Review Board, ARB-2026-08 (2026-08-27) and ARB-2026-09 (2026-09-08)
Supersedes	v3.3 (2026-06-30)
Review cycle	Semi-annual, or after any Sev1 or Sev2 incident

Table 2.0. Document control for the System Architecture Design.

1. Purpose, scope and principles

The Payment Risk & Fraud Intelligence Platform analyzes every card and wallet payment routed through Halcyon Payments and returns a decision within a strict latency budget: approve, request additional authentication, send for manual review, or decline. The platform also supports fraud operations through case management, feeds confirmed outcomes back into model training, and provides the monitoring needed to operate machine learning models safely in production.

This document describes the components, data flows, deployment, scaling behavior and failure modes of the platform. It is written for engineers who build and operate the platform, for architects who review changes, and for auditors who assess resilience. Model details are in Part 3, security controls in Part 4, operational procedures in Part 5, incident history in Part 6, and disaster recovery in Part 7.

1.1 Architecture principles

1. **The scoring path must degrade gracefully.** Every dependency on the scoring path has a defined fallback, so that no single failure stops decisions from being made.
2. **The request path is synchronous and independent of Kafka.** Scoring never waits for asynchronous infrastructure.
3. **Derived data is rebuildable.** The online feature store is a cache of derived data; the system of record is the event log and the offline store.
4. **Every decision is reproducible.** Each decision stores the model version, rule set version, threshold set version and the exact feature snapshot used.
5. **No full card numbers.** PRFIP receives only tokens and never processes a PAN, keeping it outside the PCI cardholder data environment.
6. **Configuration over code for business thresholds.** Thresholds, rule sets and routing are versioned configuration with governance approval.
7. **Explicit budgets.** Every downstream call has its own time budget so that the 250 ms overall deadline can be met.
8. **Observable by default.** Every component emits metrics, structured logs and traces using the shared conventions.

1.2 Scope

In scope: the scoring path, streaming feature computation, the online and offline feature stores, model serving, rules and decisioning, case management, the label feedback loop, model registry and monitoring, audit logging and observability. Out of scope: the Payment Gateway and Token Vault (owned by the Payment Gateway team), issuer authorization, settlement, and merchant onboarding. Interfaces to those systems are described where they affect the platform.

2. System overview

2.1 Context

PRFIP sits between the Payment Gateway and the card networks. When a merchant submits a payment, the Payment Gateway calls the Fraud Scoring API before it forwards the authorization to the issuer. The platform answers within 250 ms. The Payment Gateway then acts on the decision. Independently and asynchronously, the Payment Gateway publishes the transaction event to Kafka, which the stream processor uses to update behavioral features for later requests.

External system	Direction	Interface	Owner
Payment Gateway	Calls PRFIP; publishes transaction events	HTTPS /v1/risk/score; Kafka payments.transactions.v1	Naomi Fischer
Token Vault	Issues card tokens (outside PRFIP)	Tokens carried in requests as card_token_id	Payment Gateway team
DeviceTrust vendor	PRFIP calls vendor	HTTPS REST via Device Intelligence Adapter	Vendor; adapter owned by Naomi Fischer
Email and phone reputation vendor	PRFIP batch loads	Daily file drop and API	Priya Raman
Merchant Platform	Provides merchant attributes	Daily extract and events	Merchant Platform team
Compliance sanctions screening	Provides screening results in request	Field in scoring request	Compliance
Chargeback processing	Provides chargeback files	Daily SFTP	Finance Operations
Microsoft Entra ID	Workforce identity for analysts and engineers	OIDC / SAML with MFA	IT Security

Table 2.1. External systems and interfaces.

2.2 Life of a scoring request

The synchronous scoring path is summarized below. All times refer to the budget allocated to each step; the sum of budgets on the critical path is smaller than the 250 ms deadline because steps 3 to 6 run in parallel.

1. The Payment Gateway sends POST /v1/risk/score with an idempotency key. The API Gateway authenticates the client, validates the schema and forwards the request (budget 5 ms).
2. The Risk Orchestrator assigns a request_id, starts the 250 ms deadline timer and computes the 32 request-only features (budget 10 ms).
3. In parallel, the Orchestrator calls the Feature Service for the six entity families (card, device, ip, email, customer, merchant), with a budget of 80 ms.

4 In parallel, the Orchestrator calls the Device Intelligence Adapter (budget 60 ms) and the Merchant Risk Service (budget 40 ms).

5 The Orchestrator assembles the 84-feature vector; features that are unavailable are encoded as null.

6 The Orchestrator calls the FraudShield Scoring Service, which returns a calibrated risk_score and the top reason contributions (budget 40 ms, typical 18 ms).

7 The Rules Engine evaluates pre- and post-score rules (budget 15 ms).

8 The Decision Service applies the threshold set for the model version and returns APPROVE, STEP_UP, REVIEW or DECLINE (budget 5 ms).

9 The Orchestrator returns the response and asynchronously publishes to risk.scores.v1, risk.decisions.v1 and features.snapshots.v1.

10 For REVIEW decisions, Case Management creates a case within 5 seconds.

Typical end-to-end latency is 64 ms (p50 35 ms, p99 under 120 ms). If any step fails or exceeds its budget the Orchestrator applies the degradation ladder described in the Failure Scenarios chapter.

2.3 Asynchronous flows

- **Feature stream:** Payment Gateway to payments.transactions.v1, Flink computes windows, results go to features.updates.v1 and then the online store.
- **Snapshot and monitoring:** the exact feature vector for each request is published to features.snapshots.v1, which feeds the offline store and the drift monitor.
- **Label loop:** analyst resolutions and chargebacks become labels in fraud.labels.v1, which feed training datasets.
- **Model lifecycle:** approved artifacts in the Model Registry are deployed by the release pipeline; deployment events are published to models.events.v1.
- **Audit:** all services publish audit events, which are written to immutable storage.

3. Component catalog

The following pages profile each of the eighteen components of the platform. Every profile uses the same fields: owner, technology, tier, purpose, inputs, outputs, dependencies, service level objective, scaling and failure behavior, followed by operational notes.

3.1 API Gateway (C01)

Attribute	Value
Owner	Marcus Lee
Technology	Kong Gateway on EKS
Tier	Tier 1
Purpose	Authenticates and rate-limits all external calls into the platform and routes /v1/risk/* and /v1/cases/* traffic.
Inputs	HTTPS requests from Payment Gateway and internal tools
Outputs	Authenticated, validated requests to Risk Orchestrator and Case Management
Dependencies	Identity provider (Microsoft Entra ID), Risk Orchestrator
Service level	99.99% availability
Scaling	Horizontal, 12 to 40 pods; HPA on CPU 60%
Failure behavior	If the gateway is unavailable, Payment Gateway applies its local fail-safe policy (see the Failure Scenarios chapter).

The gateway terminates mutual TLS for partner clients, validates OAuth 2.0 client-credentials tokens, and enforces per-client rate limits (default 6,000 requests per second per client_id). Request bodies are schema-validated against the published API contract; malformed requests are rejected with HTTP 400 before reaching the scoring path. A 300 ms upstream timeout is configured so that the gateway never holds a connection longer than the client-side deadline of 400 ms. Kong plugins add the X-Request-Id header and propagate W3C trace context to all downstream services.

3.2 Risk Orchestrator (C02)

Attribute	Value
Owner	Marcus Lee
Technology	Java 21, Spring Boot, gRPC
Tier	Tier 1
Purpose	Coordinates one scoring request: gathers request features, calls Feature Service, Device Intelligence Adapter and Merchant Risk Service in parallel, calls FraudShield, then calls Rules Engine and Decision Service.
Inputs	Scoring requests from API Gateway
Outputs	Decision response; events to risk.scores.v1 and risk.decisions.v1
Dependencies	Feature Service, FraudShield Scoring Service, Rules Engine, Decision Service, Device Intelligence Adapter
Service level	Server-side deadline 250 ms; p99 end-to-end 120 ms
Scaling	Horizontal, 30 to 120 pods; HPA on in-flight requests
Failure behavior	Enforces the 250 ms deadline; on any downstream failure it applies the degradation ladder in the Failure Scenarios chapter.

The Orchestrator owns the request deadline and the degradation ladder. Each downstream call has its own budget: Feature Service 80 ms, Device Intelligence Adapter 60 ms, Merchant Risk Service 40 ms, FraudShield 40 ms, Rules Engine 15 ms. Calls that do not depend on each other run in parallel. If the

Feature Service budget is exceeded the Orchestrator switches that request to the reduced-feature model FraudShield v4.2-RF; if the overall 250 ms deadline is exceeded it returns a rules-only decision with degraded=true. Every request writes a feature snapshot to features.snapshots.v1 so that training-serving skew can be measured.

3.3 Feature Service (C03)

Attribute	Value
Owner	Priya Raman
Technology	Java 21, Lettuce Redis client
Tier	Tier 1
Purpose	Serves precomputed behavioral and entity features from the online store to the Orchestrator at scoring time.
Inputs	Entity keys (card_token_id, device_id, ip_address, email_hash, customer_id, merchant_id)
Outputs	Feature vectors grouped by entity family
Dependencies	Feature Store Online (Redis Cluster)
Service level	p99 25 ms; 99.95% availability
Scaling	Horizontal, 20 to 80 pods; Redis connection pool 480 per pod (raised from 200 after INC-2026-041)
Failure behavior	If unavailable, Orchestrator uses the reduced-feature model (Failure Scenario FS-1).

The Feature Service accepts a list of entity keys and returns the latest feature values for each entity family: card, device, ip, email, customer and merchant. Version 4.2 of FraudShield requires six grouped lookups per request, one per entity family. Lookups use pipelined MGET commands to keep the number of Redis round trips small. Each returned value carries an event timestamp so that the Orchestrator can compute feature age and mark stale values as missing when age exceeds 6 hours. Missing values are encoded as null and handled natively by the model.

3.4 Feature Store Online (C04)

Attribute	Value
Owner	Priya Raman
Technology	Amazon ElastiCache for Redis (cluster mode), 12 shards
Tier	Tier 1
Purpose	Low-latency key-value store holding the latest value of each streaming and batch feature per entity.
Inputs	features.updates.v1 (Flink), hourly and daily batch loads
Outputs	Feature values by key
Dependencies	Stream Processor, Snowflake offline store (batch backfill)
Service level	p99 read 3 ms; 99.99% availability
Scaling	12 shards, 1 primary and 2 replicas each; memory alert at 80%
Failure behavior	Shard failover takes about 30 seconds; reads served by replicas meanwhile.

The online store is a cache of derived data, not a system of record: every value can be rebuilt by replaying payments.transactions.v1 through the stream processor and by re-running batch feature jobs. TTL is 6 hours for streaming features and 36 hours for batch features. Keys follow the pattern feat:{family}:{entity_id}. Memory is sized for 1.8x current key cardinality. Replication to the disaster recovery region is asynchronous with a target lag under 5 minutes.

3.5 Feature Store Offline (C05)

Attribute	Value
Owner	Priya Raman
Technology	Snowflake and Amazon S3 (Parquet, Iceberg tables)
Tier	Tier 3
Purpose	Historical feature values and point-in-time correct training datasets.
Inputs	features.snapshots.v1, transaction and label history
Outputs	Training datasets, backfills, analytics
Dependencies	Data Lake, Label and Feedback Service
Service level	Not on the scoring path; daily freshness
Scaling	Elastic warehouse; XL for training extracts
Failure behavior	Outage delays model training and batch feature refresh but does not affect scoring.

The offline store keeps feature snapshots exactly as they were served, enabling point-in-time joins with fraud labels and measurement of training-serving skew. Snapshots are retained 25 months. Access to snapshot tables containing transaction-level features requires the Confidential-tier role described in the Security Architecture.

3.6 FraudShield Scoring Service (C06)

Attribute	Value
Owner	Sarah Johnson
Technology	Python 3.11, LightGBM, KServe on EKS
Tier	Tier 1
Purpose	Hosts FraudShield model versions and returns a calibrated fraud probability for each request.
Inputs	84-feature vector from Orchestrator
Outputs	risk_score between 0 and 1, model_version, top reason contributions
Dependencies	Model Registry (artifact load at startup)
Service level	p99 inference 18 ms; 99.95% availability
Scaling	Horizontal, 16 to 64 pods; CPU-based inference; 2 vCPU per pod
Failure behavior	If unhealthy, Orchestrator falls back per Failure Scenario FS-2.

The service loads model artifacts from the Model Registry at startup and refuses to start if the artifact checksum does not match the registry entry. Production serves FraudShield v4.2 (84 features) and, in shadow mode only, FraudShield v5.0 (112 features); shadow scores are logged but never returned to

callers. A reduced-feature variant, FraudShield v4.2-RF, is trained on the 32 request-only features and is used when the Feature Service is unavailable. Isotonic calibration is applied so that a score of 0.85 corresponds to an empirical fraud precision of roughly 0.85 on the validation window.

3.7 Rules Engine (C07)

Attribute	Value
Owner	Tom Alvarez
Technology	Drools on JVM, rules in Git
Tier	Tier 1
Purpose	Applies deterministic business and compliance rules before and after model scoring.
Inputs	Request features and model score
Outputs	Rule hits and rule-based decision overrides
Dependencies	None on the request path (rules are loaded in memory)
Service level	p99 10 ms; 99.99% availability
Scaling	Embedded in Orchestrator pods; rule sets hot-reloaded
Failure behavior	Rules-only mode is the last step of the degradation ladder.

Rules are versioned in Git and promoted through the Change Management Procedure. Hard-decline rules cover sanctions screening results, blocked BINs and blocked devices. Velocity guard rules decline transactions when a single card exceeds 20 transactions in one hour. Fraud Operations may propose new rules; every rule requires an owner, an expiry date and a documented false-positive estimate before approval.

3.8 Decision Service (C08)

Attribute	Value
Owner	Marcus Lee
Technology	Java 21
Tier	Tier 1
Purpose	Combines model score, rule hits and business context into a final decision: APPROVE, STEP_UP, REVIEW or DECLINE.
Inputs	risk_score, rule hits, transaction context
Outputs	decision, decision_reason_codes
Dependencies	Rules Engine
Service level	p99 5 ms
Scaling	Embedded in Orchestrator pods
Failure behavior	Falls back to conservative static thresholds if configuration cannot be loaded.

Thresholds are configuration, not code, and are versioned per model version. For FraudShield v4.2 the approved thresholds are: DECLINE at or above 0.85, REVIEW from 0.60 to below 0.85, STEP_UP from 0.50 to below 0.60 for e-commerce transactions without 3-D Secure authentication, and APPROVE otherwise. Threshold changes require Model Governance Committee approval.

3.9 Device Intelligence Adapter (C09)

Attribute	Value
Owner	Naomi Fischer
Technology	Go service wrapping third-party DeviceTrust API
Tier	Tier 2
Purpose	Retrieves device trust signals from the external DeviceTrust vendor and caches them.
Inputs	device_fingerprint_hash, ip_address
Outputs	device_trust_score, emulator and root flags, IP reputation
Dependencies	DeviceTrust vendor API (external)
Service level	p99 45 ms including cache hits; vendor SLA 99.9%
Scaling	Horizontal, 8 to 24 pods; 15-minute cache
Failure behavior	If the vendor is unavailable the adapter returns null signals; the model handles nulls.

Vendor responses are cached for 15 minutes per device fingerprint. A circuit breaker opens after five consecutive vendor timeouts and stays open for 30 seconds. While open, device-derived features are null and the request is marked with reason code DEVICE_SIGNAL_MISSING for later analysis.

3.10 Merchant Risk Service (C10)

Attribute	Value
Owner	Sarah Johnson
Technology	Python FastAPI
Tier	Tier 2
Purpose	Provides merchant-level risk attributes, including merchant_risk_score, computed from merchant history.
Inputs	Merchant chargeback, refund and volume history
Outputs	merchant_risk_score and merchant risk tier
Dependencies	Feature Store Offline
Service level	p99 30 ms
Scaling	Horizontal, 6 to 16 pods
Failure behavior	Stale values are used if the daily refresh fails; alert raised after 36 hours.

merchant_risk_score is calculated by a daily batch job from 90 days of merchant chargeback rate, refund ratio, transaction volume growth and merchant age, then refreshed hourly for merchants with material new activity. The score is stored in the online store and returned to the Feature Service; it is not computed on the request path.

3.11 Stream Processor (C11)

Attribute	Value
Owner	Wei Zhang
Technology	Apache Flink 1.19 on EKS
Tier	Tier 1
Purpose	Computes streaming velocity and behavioral features from transaction events.
Inputs	payments.transactions.v1, fraud.labels.v1
Outputs	features.updates.v1
Dependencies	Kafka (MSK), Schema Registry
Service level	Feature freshness p95 under 60 seconds
Scaling	Parallelism 48; checkpoints every 60 seconds to S3
Failure behavior	Failure leaves online features stale; freshness alert pages the on-call SRE (added after INC-2026-057).

The job maintains keyed state for card, device, ip, email, customer and merchant entities and emits updated aggregates for 1-hour, 24-hour, 7-day and 30-day windows. Device velocity features such as device_velocity_24h are computed here. Deserialization failures are routed to a dead-letter topic; before INC-2026-057 they caused the job to restart in a loop.

3.12 Event Bus (C12)

Attribute	Value
Owner	Wei Zhang
Technology	Amazon MSK (Kafka 3.6), Confluent Schema Registry
Tier	Tier 1
Purpose	Durable event backbone between the Payment Gateway, scoring path, stream processing, storage and monitoring.
Inputs	Events from all producers
Outputs	Ordered, durable topics
Dependencies	AWS multi-AZ
Service level	99.95% availability
Scaling	9 brokers across 3 AZs; replication factor 3; min.insync.replicas 2
Failure behavior	Producers buffer locally for up to 30 seconds; scoring path does not depend on Kafka being available.

The scoring request path is synchronous and does not wait on Kafka. Kafka carries the asynchronous flows: transaction events, scores, decisions, feature updates, labels and audit events. Schema Registry enforces BACKWARD_TRANSITIVE compatibility on all PRFIP subjects since 9 September 2026 (REL-2026-036).

3.13 Case Management Service (C13)

Attribute	Value
Owner	Tom Alvarez
Technology	Java 21, PostgreSQL 15 (Aurora)
Tier	Tier 2
Purpose	Creates and tracks manual review cases and analyst decisions.
Inputs	risk.decisions.v1 (REVIEW decisions), analyst actions
Outputs	Case records, resolutions, labels
Dependencies	Aurora PostgreSQL, Label and Feedback Service
Service level	99.9% availability; case created within 5 seconds of REVIEW decision
Scaling	Horizontal, 6 to 20 pods
Failure behavior	If unavailable, review decisions are queued in Kafka and cases are created on recovery.

Cases are assigned to queues by amount band and merchant segment. The target time to first analyst touch is 15 minutes for high-value cases and 45 minutes otherwise. Analysts resolve cases as CONFIRMED_FRAUD, LEGITIMATE or INCONCLUSIVE; resolutions are published to fraud.labels.v1.

3.14 Label and Feedback Service (C14)

Attribute	Value
Owner	Sarah Johnson
Technology	Python, Airflow
Tier	Tier 2
Purpose	Consolidates chargebacks, analyst resolutions and issuer fraud reports into training labels.
Inputs	Chargeback files, case resolutions
Outputs	fraud.labels.v1, labeled training tables
Dependencies	Case Management Service, Snowflake
Service level	Daily; chargeback labels mature after 120 days
Scaling	Airflow DAG, 3 workers
Failure behavior	Delays labeling; does not affect scoring.

Because chargebacks arrive weeks after a transaction, the service tags each label with a maturity level. Training datasets only include transactions older than 120 days as fully matured; recent months use analyst-confirmed labels with a documented bias adjustment.

3.15 Model Registry (C15)

Attribute	Value
Owner	Sarah Johnson
Technology	MLflow on EKS, S3 artifact store
Tier	Tier 2
Purpose	Stores model artifacts, model cards, validation status and lifecycle stage.
Inputs	Training pipeline outputs
Outputs	Approved artifacts for deployment
Dependencies	S3, Aurora PostgreSQL
Service level	99.5% availability
Scaling	2 pods
Failure behavior	Existing deployments continue; new deployments blocked.

Each registered model version carries lifecycle status (Development, Staging, Shadow, Production, Archived, Rejected), the validation report ID, the approving committee meeting and the effective date. Only a version in Production status can be served to callers; Shadow versions can be loaded only by the shadow route.

3.16 Model Monitoring and Drift Service (C16)

Attribute	Value
Owner	Sarah Johnson
Technology	Python, Evidently, Prometheus
Tier	Tier 2
Purpose	Measures score and feature drift, precision proxies and decision rates.
Inputs	risk.scores.v1, features.snapshots.v1, fraud.labels.v1
Outputs	Metrics and alerts
Dependencies	Kafka, Prometheus
Service level	Metrics every 5 minutes; PSI daily and hourly
Scaling	2 pods
Failure behavior	Loss of monitoring is itself alerted after 15 minutes without metrics.

The service computes population stability index (PSI) for the score and for the 20 most important features against a rolling 28-day reference window, and estimates precision at the decline threshold from analyst-reviewed cases. Thresholds and rollback conditions are defined in the Model Monitoring Specification and repeated in the Production Runbook.

3.17 Audit Logging Service (C17)

Attribute	Value
Owner	Elena Petrova
Technology	Java, Kafka consumer, S3 Object Lock
Tier	Tier 2
Purpose	Writes an immutable audit trail of decisions, configuration changes and privileged access.
Inputs	audit.events.v1
Outputs	Immutable audit records
Dependencies	S3 Object Lock (compliance mode)
Service level	Pipeline lag under 5 minutes
Scaling	2 pods
Failure behavior	Events are retained in Kafka for 14 days if the service is down.

Every decision is auditable: request_id, model_version, rule set version, threshold set version, feature snapshot reference and decision reason codes. Audit records are retained 7 years and cannot be modified or deleted within that period.

3.18 Observability Stack (C18)

Attribute	Value
Owner	Daniel Okafor
Technology	Prometheus, Grafana, OpenTelemetry, PagerDuty
Tier	Tier 2
Purpose	Collects metrics, logs and traces and routes alerts to on-call engineers.
Inputs	Metrics, traces and logs from all components
Outputs	Dashboards and pages
Dependencies	Prometheus, Loki, Tempo, PagerDuty
Service level	99.9% availability
Scaling	HA pairs
Failure behavior	Loss of dashboards does not affect scoring; Prometheus outage pages the SRE lead.

Golden-signal dashboards exist for each Tier 1 component. Alerts are classified Sev1 to Sev4; Sev1 and Sev2 page the on-call engineer through PagerDuty and open an incident channel automatically.

4. Event streaming and data architecture

4.1 Kafka topics

Kafka (Amazon MSK) runs nine brokers across three availability zones with replication factor 3 and min.insync.replicas of 2. Producers use idempotent writes. All schemas are registered in Confluent Schema Registry; since 2026-09-09 every PRFIP subject enforces BACKWARD_TRANSITIVE compatibility, so a producer cannot register a schema that existing consumers cannot read (REL-2026-036, ADR-012).

Topic	Producer	Consumers	Partitions	Retention	Key	Schema	Classification
payments.transactions.v1	Payment Gateway	Stream Processor, Data Lake sink	48	7 days	card_token_id	1.7	Confidential
payments.transactions.dlq.v1	Stream Processor	SRE (manual triage)	6	30 days	card_token_id	1.7	Confidential
risk.scores.v1	FraudShield Scoring Service	Decision archive, Monitoring, Data Lake	48	14 days	request_id	2.3	Confidential
risk.decisions.v1	Risk Orchestrator	Case Management, Data Lake, Audit	48	14 days	request_id	2.1	Confidential
features.updates.v1	Stream Processor	Online store writer	96	3 days	entity_key	1.4	Confidential
features.snapshots.v1	Risk Orchestrator	Offline store, Monitoring	48	14 days	request_id	1.5	Confidential
devices.signals.v1	Device Intelligence Adapter	Stream Processor, Data Lake	12	7 days	device_id	1.2	Confidential
merchants.risk-updates.v1	Merchant Risk Service	Online store writer	6	7 days	merchant_id	1.1	Internal
fraud.cases.v1	Case Management Service	Data Lake, Reporting	12	30 days	case_id	1.3	Confidential
fraud.labels.v1	Label and Feedback Service	Stream Processor, Training pipeline	12	30 days	transaction_id	1.2	Confidential
models.events.v1	Model Registry, Deployment tooling	Monitoring, Audit	3	90 days	model_version	1.0	Internal
audit.events.v1	All services	Audit Logging Service	24	14 days	actor_id	1.6	Confidential

Table 4.1. Kafka topics.

4.2 Data stores

Store	Technology	Contents	Retention	Classification
Online feature store	ElastiCache Redis cluster, 12 shards	Latest feature values per entity	6 h (streaming), 36 h (batch) TTL	Confidential
Offline feature store	Snowflake and S3 Iceberg	Feature snapshots, labels, training sets	25 months	Confidential
Transaction data lake	S3 Parquet	Raw events, scores, decisions	13 months hot, 7 years archive	Confidential
Case database	Aurora PostgreSQL 15	Cases, notes, resolutions	7 years	Confidential
Model registry	MLflow with S3	Artifacts, model cards, lifecycle status	Life of model plus 7 years	Confidential
Audit store	S3 Object Lock (compliance mode) with immutable copy in Azure Blob Storage	Decision and access audit trail	7 years	Confidential
Metrics and logs	Prometheus, Loki, Tempo	Operational telemetry	13 months metrics, 90 days logs	Internal

Table 4.2. Data stores and retention.

4.3 Principal data flows

ID	Flow	Path	Latency target
DF-1	Scoring request	API Gateway to Orchestrator to Feature Service, FraudShield, Rules, Decision	under 120 ms p99
DF-2	Feature computation	Kafka transactions to Flink to features.updates.v1 to Redis	freshness p95 under 60 s
DF-3	Decision archival	Orchestrator to risk.scores.v1 and risk.decisions.v1 to data lake	under 5 minutes
DF-4	Feature snapshot	Orchestrator to features.snapshots.v1 to offline store	under 15 minutes
DF-5	Label feedback	Case resolutions and chargebacks to fraud.labels.v1 to training tables	daily
DF-6	Model deployment	Registry to KServe, deployment event to models.events.v1	per release
DF-7	Audit	All services to audit.events.v1 to immutable store	under 5 minutes

Table 4.3. Principal data flows.

5. Infrastructure and deployment

5.1 Cloud services

Amazon Web Services is the primary cloud. Production runs in us-east-1 across three availability zones; the warm standby runs in us-west-2. Microsoft Azure is used for workforce identity (Microsoft Entra ID) and for a secondary immutable copy of the audit archive, which protects against a single-cloud failure of audit evidence.

Service	Use	Configuration
Amazon EKS	Container platform for all services	Kubernetes 1.30; dedicated node groups for Tier 1 workloads

Service	Use	Configuration
Amazon MSK	Kafka	9 brokers, kafka.m7g.2xlarge, 3 AZs, tiered storage off
ElastiCache for Redis	Online feature store	Cluster mode, 12 shards, 1 primary and 2 replicas each
Amazon S3	Data lake, checkpoints, artifacts	Versioning on; KMS customer-managed keys; Object Lock for audit
Amazon Aurora PostgreSQL	Case database, MLflow backend	Multi-AZ; cross-region read replica in us-west-2
Snowflake on AWS	Offline analytics and training	Enterprise edition; private connectivity
AWS KMS	Encryption keys	Separate keys per data domain; annual rotation
AWS PrivateLink	Connectivity to Payment Gateway VPC	No public path for scoring traffic
Microsoft Entra ID	Workforce SSO and MFA	Conditional access; privileged role activation with approval
Azure Blob Storage	Secondary immutable audit copy	Immutable policy, 7 years

Table 5.1. Cloud services.

5.2 Environments

Environment	Purpose	Data	Capacity vs production
Development	Feature and model development	Synthetic and sampled, tokenized data	10%
Staging	Integration and load testing	Replayed production traffic pattern, tokenized	60% of peak (raised to 120% of trailing peak for Tier 1 releases after INC-2026-041)
Production (us-east-1)	Live traffic	Live	100%
Disaster recovery (us-west-2)	Warm standby	Replicated	50% steady state; scales to 100% on failover

Table 5.2. Environments.

5.3 Network zones

Workloads run in separate Kubernetes namespaces and network policies default to deny. The API Gateway is the only component reachable from the Payment Gateway VPC, through PrivateLink. Service-to-service traffic is encrypted with mutual TLS using certificates rotated every 24 hours. Data-plane components (Redis, Aurora, MSK) accept connections only from their owning services. Engineers reach production only through a bastion with session recording and time-limited privileged access.

6. Performance and scalability

6.1 Latency budget

Step	Budget (ms)	Typical (ms)	Notes
API Gateway	5	2	Authentication, validation
Request feature computation	10	4	32 request-only features
Feature Service lookup	80 (typical 12)	12	Six grouped lookups, pipelined; runs in parallel with vendor calls
Device Intelligence Adapter	60	8	15-minute cache; vendor call on miss
Merchant Risk Service	40	6	Reads precomputed score
FraudShield inference	40 (typical 18)	18	p99 18 ms for v4.2
Rules Engine	15	4	In-process
Decision Service	5	1	Configuration lookup
Overall server-side deadline	250	64	p99 under 120 ms; hard deadline 250 ms
Client-side timeout (Payment Gateway)	400	n/a	Local fail-safe policy applies after 400 ms

Table 6.1. Latency budget for /v1/risk/score.

6.2 Capacity

Normal peak traffic is 3,900 requests per second. The highest observed peak was 4,800 requests per second on 2026-08-11. The platform is designed and load tested for 7,500 requests per second. Since INC-2026-041 every Tier 1 release must be load tested at 120% of the trailing 30-day peak using the release's actual call pattern.

Component	Normal pods	Max pods	Scaling signal	Design capacity
API Gateway	12	40	CPU 60%	10,000 rps
Risk Orchestrator	30	120	In-flight requests	7,500 rps
Feature Service	20	80	CPU and pool utilization	7,500 rps
FraudShield Scoring Service	16	64	CPU 65%	8,000 rps
Device Intelligence Adapter	8	24	CPU	7,500 rps
Stream Processor	48 slots	96 slots	Consumer lag	25,000 events/s

Table 6.2. Scaling configuration.

Redis capacity is governed by connections as well as memory. Each Feature Service pod holds a connection pool of 480 connections (raised from 200 in SS-2.14.1); at 80 pods this is 38,400 connections against a cluster limit sized for 60,000. Pool utilization above 75% raises alert PRFIP-FS-POOL-001.

7. Failure scenarios and degradation

7.1 The degradation ladder

The Risk Orchestrator applies a fixed degradation ladder so that the platform continues to produce decisions when components fail. Each level trades accuracy for availability, and every degraded decision is marked degraded=true with a decision_source and reason code so that its effect can be measured.

Level	Condition	Behavior	decision_source	Expected effect
0	Normal	FraudShield v4.2 with 84 features	MODEL	Baseline
1	Vendor or enrichment signals missing	Missing values sent as null; model handles nulls	MODEL	Small loss of accuracy; reason code DEVICE_SIGNAL_MISSING
2	Feature Service unavailable or over its 80 ms budget	FraudShield v4.2-RF scores using only the 32 request-only features; stricter thresholds (DECLINE at or above 0.92, REVIEW 0.75 to below 0.92, no step-up)	MODEL_REDUCED_FEATURES	Recall at 1% false positive rate falls from 0.78 to 0.52; more manual review
3	FraudShield unavailable or 250 ms deadline exceeded	Rules-only decision: rules R-004, R-007, R-009 (amount above USD 2,000 goes to REVIEW) and R-010; risk_score null	RULES_ONLY	Higher fraud exposure; response still HTTP 200 with degraded=true
4	Platform unreachable or no response within 400 ms	Payment Gateway local policy: approve below USD 100 for cached trusted customers, otherwise hold for manual review	Gateway	Operated by the Payment Gateway team

Table 7.1. Degradation ladder.

Answer to a common question: When the Feature Service becomes unavailable, the Risk Orchestrator switches to Level 2 of the ladder: it scores with the reduced-feature model FraudShield v4.2-RF using the 32 request-only features and stricter thresholds. Decisions are marked degraded=true with reason code FEATURES_UNAVAILABLE, alert PRFIP-FS-AVAIL-001 pages the on-call engineer, and Runbook procedure RB-02 applies. Scoring does not stop.

7.2 Scenario catalog

7.3 FS-1: Feature Service unavailable

Attribute	Value
Trigger	All Feature Service pods unhealthy, or every lookup exceeds the 80 ms budget
Detection	Health check fails 3 times (PRFIP-FS-AVAIL-001); degraded decision rate above 1% (PRFIP-DEG-001)
System behavior	Level 2: FraudShield v4.2-RF with stricter thresholds; degraded=true; reason FEATURES_UNAVAILABLE
Impact	Accuracy falls; manual review volume rises about 25 to 40%; no customer-visible errors
Runbook	RB-02

7.4 FS-2: FraudShield Scoring Service unavailable

Attribute	Value
Trigger	Pods not ready, or model artifact failed checksum
Detection	PRFIP-FSVC-001 (Sev1)
System behavior	Level 3: rules-only decisions; risk_score null; TIMEOUT or MODEL_UNAVAILABLE reason
Impact	Higher fraud exposure; review and decline mix changes; fraud losses rise
Runbook	RB-01

7.5 FS-3: Redis shard failover

Attribute	Value
Trigger	Primary of one shard fails
Detection	Redis failover events; replica lag alert
System behavior	Reads served by replicas in about 30 seconds; a small share of lookups time out and use null features
Impact	None to minor
Runbook	RB-04

7.6 FS-4: Redis connection pool exhaustion

Attribute	Value
Trigger	Pool utilization reaches 100% in Feature Service pods; requests queue
Detection	PRFIP-FS-POOL-001, PRFIP-FS-LAT-001, PRFIP-LAT-002
System behavior	Feature Service exceeds its 80 ms budget; Orchestrator falls to Level 2 for affected requests; deadline exceeded rate rises
Impact	This is the failure mode of INC-2026-041
Runbook	RB-04

7.7 FS-5: Kafka availability zone loss

Attribute	Value
Trigger	One AZ unavailable
Detection	Broker under-replicated partitions; producer errors
System behavior	Cluster continues with min.insync.replicas of 2; scoring path unaffected; feature freshness may lag
Impact	Freshness slightly worse
Runbook	RB-03

7.8 FS-6: Stream processor failure or restart loop

Attribute	Value
Trigger	Job cannot deserialize records or loses state
Detection	PRFIP-FLK-001, PRFIP-FLK-002, PRFIP-FS-FRESH-001, PRFIP-KFK-002
System behavior	Online features stay at last values (TTL 6 hours); streaming features become stale; scoring continues
Impact	Model input drift; possible missed fraud (INC-2026-057)
Runbook	RB-03

7.9 FS-7: DeviceTrust vendor outage

Attribute	Value
Trigger	Vendor API timeouts
Detection	Circuit breaker opens after 5 consecutive timeouts
System behavior	Device features null; reason DEVICE_SIGNAL_MISSING; Level 1
Impact	Minor accuracy loss
Runbook	RB-02

7.10 FS-8: Merchant Risk Service stale

Attribute	Value
Trigger	Daily job fails
Detection	Alert after 36 hours without refresh
System behavior	Last known merchant_risk_score used
Impact	Minor
Runbook	RB-08

7.11 FS-9: Case Management unavailable

Attribute	Value
Trigger	Aurora or service failure
Detection	PRFIP-CASE-001
System behavior	REVIEW decisions continue; case events queue in Kafka; created on recovery; analysts work from emergency queue export
Impact	Review delay
Runbook	RB-09

7.12 FS-10: Model Registry unavailable

Attribute	Value
Trigger	MLflow or S3 outage
Detection	Registry health alert
System behavior	Running pods unaffected; restarting pods load the last-known-good artifact cached on node volumes; new deployments blocked
Impact	None
Runbook	RB-10

7.13 FS-11: Regional failure of us-east-1

Attribute	Value
Trigger	Loss of primary region
Detection	Multiple Sev1 alerts; external monitor
System behavior	Failover to us-west-2; RTO 30 minutes, RPO 5 minutes; online store rebuilt from replicated data and Kafka replay
Impact	Up to 30 minutes at Level 4 for some traffic
Runbook	RB-07

7.14 FS-12: Faulty model deployment

Attribute	Value
Trigger	Calibration or drift problem after release
Detection	PRFIP-DRF-002, PRFIP-PRC-001, PRFIP-INF-002
System behavior	Rollback according to criteria in Part 3 and RB-05
Impact	Depends on severity
Runbook	RB-05

7.15 FS-13: API Gateway unavailable

Attribute	Value
Trigger	Kong pods unavailable
Detection	External synthetic check; Payment Gateway errors
System behavior	Level 4: Payment Gateway local policy
Impact	Reduced fraud protection
Runbook	RB-06

7.16 FS-14: Audit pipeline delayed

Attribute	Value
Trigger	Audit consumer down
Detection	PRFIP-AUD-001
System behavior	Decisions continue; events retained in Kafka 14 days; backfilled on recovery
Impact	None operationally; compliance evidence delayed
Runbook	RB-03

7.17 What is not degraded

Two behaviors never degrade. First, hard-decline rules for sanctions matches (R-001) and blocked BINs and devices (R-002, R-003) always run, including in rules-only mode. Second, audit events are always emitted; if the audit pipeline is unavailable the events are retained in Kafka.

8. Architecture decisions

Architecture decisions are recorded as ADRs and approved by the Architecture Review Board. The status of each record matters: APPROVED decisions are in force, SUPERSEDED decisions describe past behavior, and PROPOSED decisions have not been approved and must not be treated as design.

ADR	Decision	Status	Date	Summary
ADR-001	Use Kafka (Amazon MSK) as the event backbone	APPROVED	2024-03-12	Approved by ARB. Chosen for durable ordering and replay, which allows rebuilding the online store from the event log.
ADR-002	Use Redis Cluster for the online feature store	APPROVED	2024-03-26	Approved. Sub-5 ms reads at required scale; data is derived and rebuildable.
ADR-003	Use Apache Flink instead of Spark Structured Streaming	APPROVED	2024-04-09	Approved. Better event-time semantics and lower state-access latency.
ADR-004	Use LightGBM for FraudShield v4 family	APPROVED	2025-11-20	Approved. Replaced XGBoost used in v3.1 for faster training and native categorical handling.
ADR-005	On scoring failure fail open and approve with 400 ms timeout	SUPERSEDED	2025-06-03	Superseded by ADR-009. Original policy approved low-risk transactions on timeout; replaced because it increased fraud exposure during outages.
ADR-006	Reduced-feature fallback model FraudShield v4.2-RF	APPROVED	2026-01-22	Approved. Model trained on the 32 request-only features used when the Feature Service is unavailable.
ADR-007	Single active region with warm standby in us-west-2	APPROVED	2025-02-18	Approved. Target RTO 30 minutes and RPO 5 minutes for the scoring path.
ADR-008	Serve models with KServe on EKS	APPROVED	2025-04-15	Approved. Standard interface and canary support.
ADR-009	Server-side deadline 250 ms; rules-only decision on deadline exceeded	APPROVED	2026-02-10	Approved. Supersedes ADR-005. Rules-only mode applies static rules R-009 and R-010 and marks the response degraded.

ADR	Decision	Status	Date	Summary
ADR-010	Active-active multi-region deployment	PROPOSED	2026-06-30	Proposed by Priya Raman. Discussed at ARB-2026-06; not approved. Deferred pending cost analysis and DR test results.
ADR-011	Per-family feature lookups in Scoring Service for v5.0 readiness	APPROVED (AMENDED)	2026-06-30	Approved at ARB-2026-06 with 14 per-family lookups. Amended 2026-08-27 after INC-2026-041 to require pipelined batching and connection pool sizing.
ADR-012	Enforce BACKWARD_TRANSITIVE schema compatibility on all PRFIP topics	APPROVED	2026-09-08	Approved after INC-2026-057. Implemented 2026-09-09 as REL-2026-036.

Table 8.1. Architecture decision log.

8.1 Decision notes

ADR-005 and ADR-009. The original policy of approving low-risk transactions when scoring failed (ADR-005, 400 ms timeout) was replaced by ADR-009 on 2026-02-10. Under ADR-009 the server-side deadline is 250 ms and the platform returns a rules-only decision. The 400 ms value remains only as the client-side timeout of the Payment Gateway.

ADR-010 (proposed). Active-active multi-region was proposed by Priya Raman at ARB-2026-06. The board discussed operating cost, data consistency of the online store across regions, and the results of quarterly disaster recovery tests, and deferred the decision. It is not part of the current architecture.

ADR-011 (approved, amended). Per-family lookups were approved to allow feature families to be added for FraudShield v5.0 without changing the Feature Service contract. After INC-2026-041 the decision was amended: the Scoring Service must batch lookups with pipelined MGET commands and connection pool sizing must be reviewed at each release.

9. Version history

Version	Date	Author	Change
v3.4	2026-09-01	Marcus Lee	Added schema compatibility enforcement, dead-letter routing, freshness alert; updated pool sizes after INC-2026-041
v3.3	2026-06-30	Marcus Lee	Added ADR-010 and ADR-011; 6 grouped lookups
v3.2	2026-02-10	Marcus Lee	Introduced 250 ms deadline (ADR-009) and reduced-feature fallback (ADR-006)
v3.1	2025-09-04	Marcus Lee	Added Case Management and Label services
v3.0	2025-02-18	Marcus Lee	Introduced warm standby region (ADR-007)

Table 9.1. Version history of the System Architecture Design.

Part 3. Fraud Model Technical Design

Attribute	Value
Document	FraudShield Model Technical Design
Version / status	v4.2.3 / CURRENT (Approved)
Effective date	2026-08-27
Model owner	Sarah Johnson, Lead Data Scientist
Independent validator	Aisha Khan, Head of Model Risk & Validation (validation report VR-2026-007)
Approval	Model Governance Committee MGC-2026-03 on 2026-03-18
Related	Real-Time Feature Engineering Design (Chapter 4), Model Serving Design (Chapter 6)

Document control for the Fraud Model Technical Design.

1. FraudShield model family and status

FraudShield is the family of supervised models that estimate the probability that a payment is fraudulent. Several versions exist at the same time in different lifecycle states. Readers should always confirm the status of a version before relying on any of its parameters. The current production model is FraudShield v4.2. Version v3.1 is archived and version v5.0 is a draft in shadow evaluation.

Version	Status	Algorithm and features	Effective	Retired	Approval	Thresholds	Note
v3.1	ARCHIVED	XGBoost, 61 features	2025-02-10	2026-04-06	MGC-2025-01 (2025-01-29)	DECLINE >= 0.90; REVIEW 0.65 to < 0.90; no step-up	Replaced by v4.2. Removed from serving registry 2026-07-06. Not a valid rollback target.
v4.0	REJECTED	LightGBM, 78 features	n/a	n/a	Not approved (VR-2025-019 failed calibration test)	n/a	Failed independent validation in November 2025; never served to production traffic.
v4.1	SUPERSEDED	LightGBM, 84 features	canary only	2026-01-14	MGC-2025-06 (canary only)	Canary thresholds only	Canary aborted 2026-01-14 after calibration drift on cross-border transactions; superseded by v4.2.
v4.2	PRODUCTION	LightGBM with isotonic calibration, 84 features	2026-04-06	current	MGC-2026-03 (2026-03-18)	DECLINE >= 0.85; REVIEW 0.60 to < 0.85; STEP_UP 0.50 to < 0.60 (e-commerce without 3DS); APPROVE otherwise	Current production model. Owner: Sarah Johnson.

Version	Status	Algorithm and features	Effective	Retired	Approval	Thresholds	Note
v5.0	DRAFT	LightGBM plus CardSeq sequence embedding, 112 features	not approved	n/a	Not approved; decision scheduled MGC-2026-10 (2026-10-15)	PROPOSED only: DECLINE >= 0.82; REVIEW 0.55 to < 0.82; STEP_UP 0.45 to < 0.55	In shadow evaluation since 2026-08-19. Scores are logged but never returned to callers.

Lifecycle status of each FraudShield version.

Current production model: FraudShield v4.2 (84 features, LightGBM with isotonic calibration) is the current production model. It was approved by the Model Governance Committee on 2026-03-18 and became effective on 2026-04-06. Its owner is Sarah Johnson. Version v5.0 is a draft in shadow evaluation since 2026-08-19 and must not be treated as production; version v3.1 is archived.

1.1 Version comparison

Attribute	v3.1 (ARCHIVED)	v4.2 (PRODUCTION)	v5.0 (DRAFT)
Algorithm	XGBoost	LightGBM with isotonic calibration	LightGBM plus CardSeq sequence embedding
Number of features	61	84	112 (84 plus 28 new)
Training window	2023-07-01 to 2024-12-31	2024-01-01 to 2025-12-31	2024-07-01 to 2026-05-31
Decline threshold	0.90	0.85	0.82 (proposed only)
Review threshold	0.65 to below 0.90	0.60 to below 0.85	0.55 to below 0.82 (proposed only)
Step-up threshold	none	0.50 to below 0.60, e-commerce without 3DS	0.45 to below 0.55 (proposed only)
Service timeout policy	400 ms, fail open (ADR-005, superseded)	250 ms deadline, rules-only (ADR-009)	250 ms deadline
ROC-AUC (out-of-time)	0.951	0.968	0.974 (shadow)
PR-AUC	0.388	0.447	0.489 (shadow)
Recall at 1% false positive rate	0.71	0.78	0.83 (shadow)
Precision at decline threshold	0.82	0.86	0.84 (shadow, at proposed threshold)
Inference latency p99	11 ms	18 ms	41 ms
Calibration error, cross-border (ECE)	0.019	0.012	0.031 (concern)
Owner	Sarah Johnson	Sarah Johnson	Sarah Johnson
Validation report	VR-2025-002	VR-2026-007	Not started

Comparison of FraudShield versions.

2. Model design

2.1 Algorithm and hyperparameters

FraudShield v4.2 is a gradient boosted decision tree ensemble trained with LightGBM. Trees handle heterogeneous tabular features, missing values and non-linear interactions natively, and they meet the inference latency budget on CPU. The raw ensemble output is calibrated using isotonic regression fitted on an out-of-time calibration window, so that the score can be interpreted as an empirical probability and thresholds keep a stable meaning across retrains.

Parameter	Value	Rationale
n_estimators	1,200 (early stopping on validation PR-AUC)	Balance of accuracy and latency
learning_rate	0.04	Stable convergence
num_leaves	96	Interaction depth without overfitting
min_data_in_leaf	400	Regularization on rare fraud patterns
feature_fraction	0.70	Reduce correlated feature dominance
bagging_fraction	0.80	Variance reduction
lambda_l2	10	Regularization
Monotone constraints	Increasing on device_velocity_1h, device_velocity_24h, card_decline_count_24h, card_txn_count_1h	Prevents implausible reversals under adversarial input
Class weighting	Legitimate transactions downsampled 1 in 25; weights restored in calibration	Training efficiency
Calibration	Isotonic regression on 2025-11-01 to 2025-12-31	Score interpretable as probability

Training configuration for FraudShield v4.2.

2.2 Training data

The v4.2 training set covers transactions from 2024-01-01 to 2025-12-31 with matured labels. Legitimate transactions were downsampled at 1 in 25, giving about 512 million rows of which 9.6 million are labeled fraud. Labels come from chargebacks (maturity 120 days), analyst-confirmed cases and issuer fraud reports. The out-of-time validation window is 2026-01-01 to 2026-02-28. Features are joined point-in-time from the offline feature store so that no feature uses information from after the transaction event time.

Dataset	Period	Rows	Fraud rows	Use
Training	2024-01-01 to 2025-10-31	438 million	8.2 million	Model fitting
Calibration	2025-11-01 to 2025-12-31	74 million	1.4 million	Isotonic calibration
Validation (out-of-time)	2026-01-01 to 2026-02-28	168 million (not downsampled)	0.56 million	Independent validation and threshold setting

Datasets used for FraudShield v4.2.

2.3 Reduced-feature model v4.2-RF

FraudShield v4.2-RF is trained on the 32 request-only features and is used when the Feature Service is unavailable (ADR-006). It achieves ROC-AUC 0.921, PR-AUC 0.301 and recall at 1% false positive rate of 0.52. Because its precision is lower, stricter thresholds apply: DECLINE at or above 0.92 and REVIEW from 0.75 to below 0.92, with no step-up band.

2.4 Decision thresholds

Thresholds convert a score into a decision. They are configuration, versioned as threshold sets, and approved by the Model Governance Committee together with the model. The set in force for FraudShield v4.2 is TS-4.2.1.

Score range	Decision	Condition
0.85 to 1.00	DECLINE	Always
0.60 to below 0.85	REVIEW	Creates a case; routed by amount band
0.50 to below 0.60	STEP_UP	E-commerce transaction without successful 3-D Secure authentication; otherwise APPROVE
Below 0.50	APPROVE	Subject to rules

Thresholds for FraudShield v4.2 (threshold set TS-4.2.1).

Proposal, not decision: Thresholds of 0.82, 0.55 and 0.45 for FraudShield v5.0 were tabled at MGC-2026-09 on 2026-09-10 as a proposal. The committee deferred the decision to MGC-2026-10 (2026-10-15). Until an approval is recorded they are not in force, and v4.2 thresholds remain the only valid production thresholds.

2.5 Explainability

Each score is accompanied by TreeSHAP contributions. The top three contributing features are mapped to the reason codes returned in decision_reason_codes, for example HIGH_DEVICE_VELOCITY when device_velocity_24h is the largest contributor. Explanations are stored with the decision for analyst use and for regulatory requests.

3. Feature engineering

FraudShield v4.2 uses 84 features in three groups: 32 request features computed by the Risk Orchestrator, 33 behavioral features computed by the stream processor, and 19 entity risk and enrichment features supplied by the Merchant Risk Service, the device vendor and batch jobs. The feature catalog follows. Data classification of transaction-level features is Confidential; static reference-data features are Internal.

3.1 Computation semantics

- Windows are trailing windows measured from the event_time of the transaction being scored; the current transaction is excluded and is added to state after scoring, when its event arrives on payments.transactions.v1.
- Late events are accepted for up to 2 minutes; later events are counted in the next window emission.
- device_velocity_24h counts prior transactions from the same device_id in the preceding 24 hours, of any outcome.
- Ratios return null when the denominator is zero.

- Values older than 6 hours (streaming) or 36 hours (batch) are treated as missing.
- merchant_risk_score originates in the Merchant Risk Service, is computed by a daily batch job with hourly incremental refresh for merchants with material new activity, and is stored in the online store.

3.2 Feature catalog for FraudShield v4.2 (84 features)

Request features (32)

Feature	Meaning	Type	Source	Window	Added in	Class
txn_amount_usd	Transaction amount converted to USD at request-time FX rate	decimal	Payment Gateway	none	v3.1	Confidential
txn_amount_local	Transaction amount in the transaction currency	decimal	Payment Gateway	none	v3.1	Confidential
currency_code	ISO 4217 currency code of the transaction	categorical	Payment Gateway	none	v3.1	Internal
payment_method_type	Payment instrument type (CARD_CREDIT, CARD_DEBIT, WALLET, BANK_TRANSFER)	categorical	Payment Gateway	none	v3.1	Confidential
entry_mode	How the payment was captured (ECOM, MOTO, POS_CHIP, POS_CONTACTLESS, POS_SWIPE)	categorical	Payment Gateway	none	v3.1	Confidential
is_card_present	True when the physical card was present	boolean	Payment Gateway	none	v3.1	Confidential
is_recurring	True for merchant-initiated recurring payments	boolean	Payment Gateway	none	v3.1	Confidential
is_3ds_authenticated	True when 3-D Secure authentication succeeded	boolean	Payment Gateway	none	v3.1	Confidential
three_ds_version	3-D Secure protocol version used	categorical	Payment Gateway	none	v4.2	Confidential
avs_result_code	Address verification result returned by the issuer	categorical	Payment Gateway	none	v3.1	Confidential
cvv_result_code	Card verification value check result	categorical	Payment Gateway	none	v3.1	Confidential
billing_shipping_country_match	True when billing and shipping countries are equal	boolean	Payment Gateway	none	v3.1	Confidential
is_ip_country_billing_match	True when IP geolocation country equals billing country	boolean	Payment Gateway	none	v3.1	Confidential
hour_of_day_local	Hour of day (0 to 23) in the cardholder billing time zone	integer	Payment Gateway	none	v3.1	Confidential
day_of_week_local	Day of week (0 to 6) in the cardholder billing time zone	integer	Payment Gateway	none	v3.1	Confidential
is_high_risk_mcc	True when the merchant category code is on the high-risk list	boolean	Reference data	none	v3.1	Internal
mcc_code	Merchant category code	categorical	Payment Gateway	none	v3.1	Internal
card_bin_country	Issuing country derived from the card BIN	categorical	BIN table	none	v3.1	Internal
card_type_bin_level	Card product level (PREPAID, STANDARD, PREMIUM, COMMERCIAL)	categorical	BIN table	none	v3.1	Internal

Feature	Meaning	Type	Source	Window	Added in	Class
issuer_merchant_country_match	True when issuer country equals merchant country	boolean	BIN table	none	v3.1	Confidential
ip_is_proxy_or_vpn	True when the IP is a known proxy or VPN exit node	boolean	Device Intelligence Adapter	none	v3.1	Confidential
ip_geo_distance_to_billing_km	Distance between IP geolocation and billing address	decimal	Payment Gateway	none	v3.1	Confidential
device_type	Device class (DESKTOP, MOBILE, TABLET, OTHER)	categorical	Payment Gateway	none	v3.1	Confidential
browser_language_matches_country	True when browser language is plausible for the billing country	boolean	Payment Gateway	none	v4.2	Confidential
email_domain_risk_tier	Static risk tier (1 to 4) of the customer email domain	categorical	Reference data	none	v3.1	Internal
email_local_part_entropy	Shannon entropy of the email local part, a signal for machine-generated addresses	decimal	Payment Gateway	none	v4.2	Confidential
is_free_email_domain	True for free webmail domains	boolean	Reference data	none	v3.1	Internal
order_item_count	Number of line items in the order	integer	Payment Gateway	none	v3.1	Confidential
order_has_digital_goods	True when the order contains digital goods	boolean	Payment Gateway	none	v4.2	Confidential
shipping_method_expedited	True when expedited shipping was selected	boolean	Payment Gateway	none	v4.2	Confidential
customer_account_age_days	Age of the merchant-side customer account at checkout, as supplied by the merchant	integer	Payment Gateway	none	v3.1	Confidential
time_since_last_txn_device_sec	Seconds since the previous transaction from the same device, computed at request time from the last-seen timestamp in the request context	integer	Payment Gateway	none	v4.2	Confidential

Request features used by FraudShield v4.2.

Behavioral (streaming) features (33)

Feature	Meaning	Type	Source	Window	Added in	Class
card_txn_count_1h	Transactions by the card in the last 1 hour	integer	Stream Processor (Flink)	1h sliding	v3.1	Confidential
card_txn_count_24h	Transactions by the card in the last 24 hours	integer	Stream Processor (Flink)	24h sliding	v3.1	Confidential
card_txn_count_7d	Transactions by the card in the last 7 days	integer	Stream Processor (Flink)	7d sliding	v3.1	Confidential
card_amount_sum_24h	Total USD amount by the card in the last 24 hours	decimal	Stream Processor (Flink)	24h sliding	v3.1	Confidential
card_amount_sum_7d	Total USD amount by the card in the last 7 days	decimal	Stream Processor (Flink)	7d sliding	v3.1	Confidential
card_distinct_merchants_24h	Distinct merchants used by the card in 24 hours	integer	Stream Processor (Flink)	24h sliding	v3.1	Confidential
card_distinct_countries_7d	Distinct merchant countries used by the card in 7 days	integer	Stream Processor (Flink)	7d sliding	v4.2	Confidential
card_decline_count_24h	Declined authorizations for the card in 24 hours	integer	Stream Processor (Flink)	24h sliding	v3.1	Confidential
card_avg_amount_30d	Average USD amount of the card over 30 days	decimal	Stream Processor (Flink)	30d sliding	v3.1	Confidential
amount_to_card_avg_ratio	Current amount divided by the 30-day average amount for the card	decimal	Stream Processor (Flink)	30d sliding	v4.2	Confidential
device_velocity_1h	Transactions from the device in the last 1 hour	integer	Stream Processor (Flink)	1h sliding	v3.1	Confidential
device_velocity_24h	Transactions from the device in the last 24 hours	integer	Stream Processor (Flink)	24h sliding	v3.1	Confidential
device_velocity_7d	Transactions from the device in the last 7 days	integer	Stream Processor (Flink)	7d sliding	v3.1	Confidential
device_distinct_cards_24h	Distinct cards seen on the device in 24 hours	integer	Stream Processor (Flink)	24h sliding	v3.1	Confidential
device_distinct_cards_7d	Distinct cards seen on the device in 7 days	integer	Stream Processor (Flink)	7d sliding	v4.2	Confidential
device_first_seen_age_days	Days since the device was first seen by the platform	integer	Stream Processor (Flink)	lifetime	v3.1	Confidential
device_decline_ratio_7d	Share of the device's transactions declined in 7 days	decimal	Stream Processor (Flink)	7d sliding	v4.2	Confidential
ip_txn_count_1h	Transactions from the IP address in 1 hour	integer	Stream Processor (Flink)	1h sliding	v3.1	Confidential
ip_txn_count_24h	Transactions from the IP address in 24 hours	integer	Stream Processor (Flink)	24h sliding	v3.1	Confidential
ip_distinct_cards_24h	Distinct cards seen from the IP in 24 hours	integer	Stream Processor (Flink)	24h sliding	v3.1	Confidential
ip_distinct_devices_24h	Distinct devices seen from the IP in 24 hours	integer	Stream Processor (Flink)	24h sliding	v4.2	Confidential

Feature	Meaning	Type	Source	Window	Added in	Class
email_txn_count_24h	Transactions using the email address in 24 hours	integer	Stream Processor (Flink)	24h sliding	v3.1	Confidential
email_distinct_cards_7d	Distinct cards used with the email in 7 days	integer	Stream Processor (Flink)	7d sliding	v4.2	Confidential
email_first_seen_age_days	Days since the email hash was first seen	integer	Stream Processor (Flink)	lifetime	v3.1	Confidential
customer_txn_count_30d	Customer transactions in 30 days	integer	Stream Processor (Flink)	30d sliding	v3.1	Confidential
customer_chargeback_count_180d	Chargebacks for the customer in 180 days	integer	Stream Processor (Flink)	180d sliding	v3.1	Confidential
customer_avg_amount_90d	Average USD amount for the customer over 90 days	decimal	Stream Processor (Flink)	90d sliding	v3.1	Confidential
merchant_txn_count_1h	Transactions at the merchant in 1 hour	integer	Stream Processor (Flink)	1h sliding	v3.1	Confidential
merchant_decline_ratio_24h	Share of merchant transactions declined in 24 hours	decimal	Stream Processor (Flink)	24h sliding	v3.1	Confidential
merchant_avg_amount_30d	Average USD amount at the merchant over 30 days	decimal	Stream Processor (Flink)	30d sliding	v3.1	Confidential
merchant_refund_ratio_30d	Refund to sales ratio at the merchant over 30 days	decimal	Stream Processor (Flink)	30d sliding	v4.2	Confidential
time_since_last_txn_card_sec	Seconds since the previous transaction on the card	integer	Stream Processor (Flink)	event	v3.1	Confidential
geo_velocity_kmh_card	Implied travel speed between the card's last two transactions	decimal	Stream Processor (Flink)	event	v4.2	Confidential

Behavioral (streaming) features used by FraudShield v4.2.

Entity risk and enrichment features (19)

Feature	Meaning	Type	Source	Window	Added in	Class
merchant_risk_score	Merchant risk score from 0 to 1 derived from 90 days of chargebacks, refunds, volume growth and merchant age	decimal	Merchant Risk Service	varies	v3.1	Confidential
merchant_chargeback_rate_90d	Merchant chargebacks divided by transactions over 90 days	decimal	Merchant Risk Service	varies	v3.1	Confidential
merchant_age_days	Days since merchant onboarding	integer	Merchant platform	varies	v3.1	Internal
merchant_category_risk_tier	Static risk tier (1 to 4) of the merchant category	categorical	Reference data	varies	v3.1	Internal
merchant_country_risk_tier	Static risk tier (1 to 4) of the merchant country	categorical	Reference data	varies	v3.1	Internal
card_bin_fraud_rate_90d	Fraud rate of the card BIN over 90 days	decimal	Feature Store Offline	varies	v3.1	Confidential
issuer_fraud_rate_90d	Fraud rate of the issuer over 90 days	decimal	Feature Store Offline	varies	v3.1	Confidential
device_trust_score	Vendor device trust score from 0 to 1000	integer	DeviceTrust vendor	varies	v4.2	Confidential
device_emulator_flag	True when the device is identified as an emulator	boolean	DeviceTrust vendor	varies	v4.2	Confidential
device_rooted_flag	True when the device is rooted or jailbroken	boolean	DeviceTrust vendor	varies	v4.2	Confidential
device_fingerprint_stability_7d	Share of the last 7 days in which the fingerprint remained unchanged	decimal	Feature Store Offline	varies	v4.2	Confidential
ip_reputation_score	IP reputation score from 0 to 100	integer	DeviceTrust vendor	varies	v3.1	Confidential
ip_asn_risk_tier	Static risk tier of the IP autonomous system	categorical	Reference data	varies	v4.2	Internal
ip_is_datacenter	True when the IP belongs to a data center range	boolean	Reference data	varies	v4.2	Internal
email_reputation_score	Email reputation score from 0 to 100	integer	Email reputation vendor	varies	v3.1	Confidential
phone_reputation_score	Phone reputation score from 0 to 100	integer	Email reputation vendor	varies	v4.2	Confidential
customer_lifetime_txn_count	Lifetime transactions of the customer	integer	Feature Store Offline	varies	v3.1	Confidential
customer_lifetime_chargeback_flag	True when the customer has ever had a chargeback	boolean	Feature Store Offline	varies	v4.2	Confidential
card_first_seen_age_days	Days since the card token was first seen	integer	Feature Store Offline	varies	v4.2	Confidential

Entity risk and enrichment features used by FraudShield v4.2.

Features added in version 4.x (23 features) are marked v4.2 in the Added in column; the remaining 61 features were already used by v3.1.

3.3 Features proposed for FraudShield v5.0 (draft)

FraudShield v5.0 adds 28 features to the 84 above, for a total of 112. These features are in shadow evaluation only. The definitions below are subject to change before approval.

Feature	Meaning	Type	Source	Window	Status	Class
session_checkout_duration_sec	Seconds between cart creation and payment submission	decimal	Merchant session events	varies	v5.0 (draft)	Confidential
session_paste_event_flag	True when card data was pasted rather than typed	boolean	Merchant session events	varies	v5.0 (draft)	Confidential
failed_login_count_24h	Failed logins on the customer account in 24 hours	integer	Merchant session events	varies	v5.0 (draft)	Confidential
account_recent_change_flag	True when email, phone or address changed in the last 24 hours	boolean	Merchant session events	varies	v5.0 (draft)	Confidential
shared_device_component_size	Size of the device-sharing graph component containing the card	integer	Graph job (daily)	varies	v5.0 (draft)	Confidential
shared_ip_component_size	Size of the IP-sharing graph component containing the card	integer	Graph job (daily)	varies	v5.0 (draft)	Confidential
merchant_network_risk_score	Risk propagated from connected merchants in the graph	decimal	Graph job (daily)	varies	v5.0 (draft)	Confidential
card_merchant_affinity_score	Familiarity of the card with the merchant based on history	decimal	Feature Store Offline	varies	v5.0 (draft)	Confidential
bin_attack_score_1h	Card-testing attack score for the BIN in the last hour	decimal	Stream Processor	varies	v5.0 (draft)	Confidential
network_token_flag	True when a network token was used instead of a card token	boolean	Payment Gateway	varies	v5.0 (draft)	Confidential
customer_session_count_24h	Sessions of the customer in 24 hours	integer	Merchant session events	varies	v5.0 (draft)	Confidential
cart_edit_count_session	Number of cart edits in the session	integer	Merchant session events	varies	v5.0 (draft)	Confidential
card_seq_embedding_00	Dimension 0 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential
card_seq_embedding_01	Dimension 1 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential
card_seq_embedding_02	Dimension 2 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential
card_seq_embedding_03	Dimension 3 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential

Feature	Meaning	Type	Source	Window	Status	Class
card_seq_embedding_04	Dimension 4 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential
card_seq_embedding_05	Dimension 5 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential
card_seq_embedding_06	Dimension 6 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential
card_seq_embedding_07	Dimension 7 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential
card_seq_embedding_08	Dimension 8 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential
card_seq_embedding_09	Dimension 9 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential
card_seq_embedding_10	Dimension 10 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential
card_seq_embedding_11	Dimension 11 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential
card_seq_embedding_12	Dimension 12 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential
card_seq_embedding_13	Dimension 13 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential
card_seq_embedding_14	Dimension 14 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential
card_seq_embedding_15	Dimension 15 of the 16-dimension CardSeq learned embedding of the card's last 20 transactions	decimal	CardSeq encoder (Feature Service, draft)	last 20 txns	v5.0 (draft)	Confidential

New features in the FraudShield v5.0 draft.

4. Real-time feature pipeline

Behavioral features are computed by an Apache Flink job that consumes payments.transactions.v1. The job keys state by entity (card, device, ip, email, customer, merchant) and maintains sliding windows of 1 hour, 24 hours, 7 days, 30 days and 180 days. Results are emitted to features.updates.v1 at most once per second per entity and written to the online store by a dedicated writer.

Property	Value
Parallelism	48 slots (96 maximum)
Checkpointing	Every 60 seconds to S3; exactly-once end to end
State backend	RocksDB, incremental checkpoints
State size	About 1.4 TB across keyed state
Allowed lateness	2 minutes
Freshness target	p95 under 60 seconds from event to online store
Deserialization errors	Routed to payments.transactions.dlq.v1 (since 2026-09-09)
Schema compatibility	BACKWARD_TRANSITIVE enforced by Schema Registry

Stream processor configuration.

Before 2026-09-09 a record that could not be deserialized caused the job to fail and restart from its last checkpoint, which produced a restart loop when a breaking schema change was published (INC-2026-057). Since then such records are sent to the dead-letter topic and the job continues. A freshness alert now pages the on-call engineer when the p95 age of streaming features exceeds 120 seconds.

5. Training and validation pipeline

1. Point-in-time dataset extraction from the offline feature store joined with matured labels.
2. Feature quality checks: null rate, range, distribution against the previous training set.
3. Model fitting with early stopping; hyperparameters recorded in MLflow.
4. Isotonic calibration on the calibration window.
5. Out-of-time evaluation on the validation window, with segment analysis by channel, region, card type and amount band.
6. Model card and validation package submitted to Model Risk & Validation for independent validation.
7. Model Governance Committee review and approval of model, thresholds and effective date.

5.1 Segment performance of FraudShield v4.2

Segment	Share of volume	Recall at 1% FPR	Precision at decline	Notes
E-commerce	71%	0.79	0.86	Primary segment
Card present	22%	0.72	0.88	Lower base fraud rate
Recurring	7%	0.81	0.90	Stable behavior
Cross-border	9%	0.74	0.83	Highest calibration error (ECE 0.012 overall; 0.021 in this segment)
New merchants (under 60 days)	4%	0.63	0.79	Cold start limitation
Amount above USD 1,000	6%	0.82	0.88	High-value focus

Segment performance on the validation window (segments overlap).

6. Model serving and inference

Models are served by KServe on EKS with CPU inference. Each pod hosts the production model and, in the shadow route, the shadow model. The inference contract accepts an 84-feature vector (112 for shadow) and returns risk_score, the model version and top contributions. Pods load artifacts at startup from the Model Registry, verify the checksum against the registry entry and then warm up with 2,000 synthetic requests before reporting ready.

Property	Value
Inference latency	p99 18 ms for v4.2; p99 41 ms for v5.0 shadow
Pod size	2 vCPU, 4 GiB
Replicas	16 normal, 64 maximum
Health check	/healthz every 5 seconds; model loaded and warm
Routing	Production route to v4.2; shadow route mirrors traffic to v5.0 and drops the response
Fallback model	FraudShield v4.2-RF loaded alongside v4.2

Serving configuration.

7. Monitoring, thresholds and rollback conditions

Monitoring is defined in the Model Monitoring Specification. Metrics are computed every five minutes (operational) or hourly and daily (statistical) by the Model Monitoring and Drift Service.

Metric	Definition	Warning	Action threshold
Score PSI	Population stability index of risk_score vs 28-day reference	Above 0.20	Above 0.25 for 2 consecutive hours
Feature PSI	PSI of each of the top 20 features	Above 0.20	Above 0.25 on any feature
Precision at decline	Share of declined transactions confirmed as fraud in analyst-reviewed sample	Below 75%	Below 65% for 24 hours
Inference latency	p99 inference time	Above 40 ms for 10 minutes	Above 60 ms for 15 minutes

Metric	Definition	Warning	Action threshold
Approval rate	Approve share vs 7-day baseline	Deviation above 2 points	Deviation above 3 points for 30 minutes
Degraded rate	Share of decisions with degraded=true	Above 0.5%	Above 1% for 10 minutes
Review rate	Share of decisions sent to REVIEW	Above 1.4%	Above 1.8%
Label-based recall	Recall on matured labels, monthly	Below 0.74	Below 0.70

Monitored metrics and thresholds.

7.1 Conditions that require rollback

A rollback of a FraudShield deployment is required when at least one of the following conditions holds, and is authorized by the Incident Commander together with the model owner or the owner's delegate:

1. Score PSI above 0.25 for two consecutive hours together with a fall in the precision proxy;
2. Estimated precision at the decline threshold below 65% for 24 hours;
3. Inference latency p99 above 60 ms for 15 minutes that cannot be resolved by scaling;
4. Approval rate deviating more than 3 percentage points from the 7-day baseline for 30 minutes without a known business cause;
5. A Critical validation finding discovered after release.

Rollback target: A rollback returns to the previous approved serving configuration of FraudShield v4.2 (the last known good release of the Scoring Service). FraudShield v3.1 is ARCHIVED and was removed from the serving registry on 2026-07-06; it is not a valid rollback target. If the fault is in the model itself, activate v4.2-RF or rules-only mode (Level 2 or 3 of the degradation ladder) and open an urgent model incident.

8. Deployment process

Stage	Gate	Approver	Minimum duration
Development	Unit tests, model card draft	Model owner	n/a
Staging	Independent validation report with no open Critical or High findings	Model Risk & Validation	n/a
Shadow	Shadow metrics reviewed; CHG approved	Model Governance Committee	14 days
Canary 5%	Latency and drift within thresholds; must include a peak traffic period	Incident Commander on duty and model owner	24 hours including a peak
Canary 25% and 50%	Same criteria	Model owner	24 hours each
Production 100%	Release Readiness Review complete	Model Governance Committee	n/a

Model promotion stages.

9. Known limitations

ID	Limitation	Effect	Mitigation
LIM-1	Cold start for merchants younger than 60 days	Lower recall (0.63)	Merchant risk tiers; rule R-006 for high-value first purchases
LIM-2	Label delay of up to 120 days	Recent months use analyst labels; possible bias	Maturity tagging; monthly re-estimation
LIM-3	Cross-border calibration error	Score less reliable for cross-border traffic	Segment monitoring; v5.0 work in progress
LIM-4	No session-behavior features	Misses bot-like checkout behavior	Planned for v5.0 (draft)
LIM-5	Dependence on vendor device signals	Degrades when the vendor is unavailable	Circuit breaker; nulls handled by model
LIM-6	Adversarial adaptation	Fraud patterns shift within weeks	Weekly monitoring; rules can be deployed within a day

Known limitations of FraudShield v4.2.

Part 4. Security Architecture

Attribute	Value
Document	Payment Data Security Architecture
Version / status	v2.6 / CURRENT (Approved)
Effective date	2026-06-15
Owner	Elena Petrova, Principal Security Architect
Data governance owner	Rahul Mehta
Standards	PCI DSS v4.0 (connected-to scope), NIST CSF, internal Data Classification Standard

Document control for the Security Architecture.

1. Scope and security principles

This part defines how the platform protects payment-related data and decisions. It covers data classification, access control, encryption, network security, logging, retention and privacy. PRFIP is designed to remain outside the PCI cardholder data environment: it never receives, processes or stores a full card number. It receives a token that stands in for the card and a small set of card attributes (BIN and last four digits) that PCI DSS permits to be stored.

- Least privilege and time-limited privileged access.
- Defense in depth with separate network zones.
- Encrypt everywhere, with keys separated by data domain.
- Every access to Confidential data is logged and reviewable.
- Security controls are code: configuration is version controlled and scanned.

2. Data classification

All platform data is classified using four tiers defined in the Data Classification Standard. Classification determines handling, encryption, access and retention.

Tier	Definition	Examples	Handling
Public	Approved for public release	Published API overview	No restriction
Internal	General business information	Aggregated dashboards, architecture documents, reference data lists	Employees only; SSO
Confidential	Customer or transaction-related data, or data whose disclosure would harm the company	Transaction-level features, risk scores, decisions, case data, device identifiers, hashed emails	Role-based access, encrypted, all access logged, no export without approval
Restricted	Data whose disclosure creates regulatory or fraud exposure	Full PAN, CVV, credentials and keys (never held by PRFIP except keys and credentials)	Named access, MFA, quarterly recertification, tokenization

Data classification tiers.

2.1 Classification of platform data

Data asset	Classification	Notes
Full card number (PAN)	Restricted	Never received by PRFIP; held only in the Token Vault (PCI CDE)
card_token_id	Confidential	Token; useless without the Token Vault
card_bin, card_last4	Confidential	Permitted for storage under PCI DSS
Transaction events (payments.transactions.v1)	Confidential	Contain tokens, hashes, IP addresses, device identifiers
Transaction-level features (online and offline store)	Confidential	Derived from transaction data; keyed by tokenized entities; snapshots retained 25 months
Static reference features (MCC, BIN country, risk tiers)	Internal	Reference lists without customer data
Risk scores and decisions	Confidential	Auditable; retained 7 years
Hashed email and phone	Confidential	SHA-256 with platform salt; no reversible storage
Case records and analyst notes	Confidential	May contain customer information
Fraud labels	Confidential	Chargeback and analyst outcomes
Model artifacts and model cards	Confidential	Intellectual property and attack-surface information
Aggregated monitoring dashboards	Internal	No entity-level data
Architecture and runbook documents	Internal	Subject to redaction of secrets
Audit logs	Confidential	Immutable

Classification of PRFIP data assets.

Answer to a common question: Transaction-level features are classified Confidential. Only static reference-data features such as MCC codes, BIN country and risk tiers are Internal, and aggregated dashboards without entity-level data are Internal.

3. Identity and access management

Workforce identity is provided by Microsoft Entra ID with multi-factor authentication and conditional access. Access to platform data and tooling is granted through role-based groups. Privileged roles are activated just in time with approval and expire after eight hours. Access reviews are quarterly for Confidential roles and semi-annual for other roles.

Role	Access	Approval	Recertification
Fraud Analyst	Case Management UI; decision and reason data for assigned cases	Fraud Operations Manager	Semi-annual
Senior Fraud Analyst	Analyst access plus rule proposals and bulk case tools	Fraud Operations Manager	Semi-annual
Data Scientist	Snowflake secure views on Confidential features; no production write	Model owner and Data Governance	Quarterly
Model Owner	Model Registry stage transitions for owned models	Head of Model Risk	Quarterly
Platform Engineer	Non-production write; production read via bastion	Head of Fraud Engineering	Quarterly

Role	Access	Approval	Recertification
SRE On-call	Production operational access with session recording	SRE Lead	Quarterly
Auditor	Read-only on audit store and configuration history	Internal Audit	Annual
Service accounts	Workload identity; no long-lived keys	Security Architect	Continuous

Roles and access.

4. Encryption and key management

Layer	Standard	Detail
External traffic	TLS 1.3	Mutual TLS for partner clients; PrivateLink from Payment Gateway
Internal traffic	mTLS	Service mesh; certificates rotated every 24 hours
Data at rest	AES-256	KMS customer-managed keys per data domain (events, features, cases, models, audit)
Key rotation	12 months	Automatic; immediate rotation on suspected compromise
Secrets	Secrets manager	No secrets in code or container images; rotation every 90 days
Hashing of identifiers	SHA-256 with salt	Salt stored in secrets manager, rotated with re-hash job annually

Encryption standards.

5. Network security

Zone	Contents	Inbound	Outbound
Edge	API Gateway	PrivateLink from Payment Gateway VPC only	Core services
Core services	Orchestrator, Feature Service, FraudShield, Rules, Decision	Edge only	Data zone; vendor egress proxy
Data	Redis, Aurora, MSK	Core services and stream processing only	Replication to DR region
Analytics	Snowflake connectivity, notebooks	Corporate network with SSO	Approved storage
Management	Bastion, CI/CD runners, observability	Corporate network with MFA	All zones (restricted)

Network zones.

6. Logging, audit and retention

Every decision records request_id, model version, rule set version, threshold set version, feature snapshot reference and reason codes. Privileged actions, access to Confidential data and configuration changes are also logged. Audit records are written to S3 with Object Lock in compliance mode and copied to an immutable Azure Blob container.

Data	Retention	Disposal
Raw transaction events	13 months hot, then 7 years archive	Cryptographic erasure at expiry
Feature snapshots	25 months	Cryptographic erasure
Risk scores and decisions	7 years	Cryptographic erasure

Data	Retention	Disposal
Case records	7 years	Deletion with certificate
Audit logs	7 years	Locked until expiry
Kafka topics	3 to 90 days by topic	Automatic
Model artifacts	Life of model plus 7 years	Deletion with certificate
Logs (Loki)	90 days	Automatic
Metrics	13 months	Automatic

Retention schedule.

7. Privacy

Emails and phone numbers are hashed before entering the platform and only the hash is stored. IP addresses are Confidential and are retained with events for 13 months. Data subject access and deletion requests are handled by Data Governance: decisions required for fraud prevention and legal obligations are retained for the statutory period, and other identifiers are erased or irreversibly hashed. Automated decisions that decline a payment carry explanations through reason codes and can be reviewed by an analyst on request.

8. Security control matrix

ID	Control	Implementation	Owner
SC-01	MFA for all workforce access	Entra ID conditional access	Elena Petrova
SC-02	Just-in-time privileged access	Privileged identity management, 8-hour expiry	Elena Petrova
SC-03	No PAN in PRFIP	Schema validation rejects PAN patterns at API Gateway	Elena Petrova
SC-04	Encryption in transit	TLS 1.3 external, mTLS internal	Marcus Lee
SC-05	Encryption at rest	KMS CMKs, AES-256	Marcus Lee
SC-06	Key rotation	Annual; on compromise	Elena Petrova
SC-07	Secrets management	Secrets manager, 90-day rotation	Daniel Okafor
SC-08	Network segmentation	Namespaces, default-deny network policies	Marcus Lee
SC-09	Vulnerability management	Critical in 72 hours internet-facing, 15 days internal	Elena Petrova
SC-10	Dependency and image scanning	CI gates on High and Critical	Marcus Lee
SC-11	Change control	All production changes through CHG process	Daniel Okafor
SC-12	Immutable audit trail	Object Lock compliance mode	Elena Petrova
SC-13	Access reviews	Quarterly for Confidential roles	Rahul Mehta
SC-14	Data minimization	Hashing of email and phone; tokens for cards	Rahul Mehta
SC-15	Row and column security on features	Snowflake secure views	Priya Raman
SC-16	Model artifact integrity	Checksum verification at load	Sarah Johnson
SC-17	Segregation of duties	Model owner cannot approve own model	Aisha Khan
SC-18	Third-party risk	Annual vendor assessment for DeviceTrust	Naomi Fischer
SC-19	Security monitoring	SIEM correlation, alert PRFIP-SEC-001	Elena Petrova
SC-20	Penetration testing	Annual external test plus quarterly internal	Elena Petrova
SC-21	Backup and recovery testing	Quarterly failover tests	Daniel Okafor
SC-22	Secure development	Code review, SAST, secrets scanning	Marcus Lee
SC-23	Data retention enforcement	Automated lifecycle policies	Rahul Mehta
SC-24	Privacy request handling	DSAR workflow with Data Governance	Rahul Mehta

Security control matrix.

9. Access to documentation and the RAG assistant

Engineering documentation for the platform is classified Internal. Sections that would reveal secrets, unmasked identifiers or exploit detail are excluded from the document set before indexing. The internal knowledge assistant must enforce the same access control as the source system: a user may receive an answer only from documents the user is entitled to read, permissions are applied at retrieval time, and every retrieval is logged. Documents with status ARCHIVED, DRAFT or PROPOSED are retrievable but must be labeled with their status in the answer.

Part 5. Production Runbook

Attribute	Value
Document	Fraud Platform Production Runbook
Version / status	v5.1 / CURRENT (Approved)
Effective date	2026-09-10
Owner	Daniel Okafor, SRE Lead
Supersedes	v5.0 (2026-08-28) and earlier
Applies to	All Tier 1 and Tier 2 components of PRFIP

Document control for the Production Runbook.

1. Using this runbook

This runbook tells the on-call engineer how to detect, diagnose and recover from production problems. Each alert names a procedure (RB-01 to RB-10). Procedures are written to be followed under pressure: read the symptoms, confirm impact, follow the numbered steps in order, and escalate when told to. If a situation is not covered, declare an incident and escalate to the SRE Lead.

1.1 On-call and escalation

Severity	Definition	Response	Notification	Update cadence
SEV-1	Scoring unavailable or majority of decisions degraded; security incident with data exposure	Page immediately; incident commander within 10 minutes	On-call SRE, SRE Lead, Head of Fraud Engineering, CISO if security, Fraud Operations	Every 15 minutes
SEV-2	Significant degradation: SLO breach, degraded decisions above 1%, stale features	Page; acknowledge within 5 minutes; commander within 15 minutes	On-call SRE, component owner, Fraud Operations	Every 30 minutes
SEV-3	Limited impact, workaround available	Ticket and Slack; respond within 1 hour in business hours	Component owner	Daily
SEV-4	Minor, no customer impact	Ticket	Component owner	As needed

Incident severity and response.

Level	Who	When
1	On-call SRE	First responder for all pages
2	Component owner on-call (Feature Platform, Streaming, Model, Case Management)	Component-specific diagnosis
3	SRE Lead (Daniel Okafor)	Any SEV-1 or SEV-2, or if unresolved after 30 minutes
4	Head of Fraud Engineering (Marcus Lee)	Any SEV-1, or customer-impacting SEV-2 over 1 hour
Business	Fraud Operations Manager (Tom Alvarez)	Any degradation that changes review volume or fraud exposure

Level	Who	When
Security	Principal Security Architect (Elena Petrova)	Any suspected unauthorized access
Model	Model owner (Sarah Johnson)	Any decision to roll back or change a model or threshold

Escalation path.

2. Service catalog and objectives

Objective	Target	Measurement
Scoring API availability	99.95% per month	Successful responses (including degraded) divided by all valid requests
Scoring API latency	p50 <= 35 ms; p99 <= 120 ms	Measured at API Gateway, 5-minute windows
Deadline exceeded rate	<= 0.5% of requests	Requests answered as TIMEOUT_RULES_ONLY
Degraded decision rate	<= 1.0% of requests	decisions with degraded=true
Feature Service latency	p99 <= 25 ms	Orchestrator-side measurement
Feature freshness	p95 <= 60 seconds	Age of streaming feature values in the online store
FraudShield inference latency	p99 <= 18 ms	Scoring Service internal measurement
Case creation	Within 5 seconds of REVIEW decision, 99% of cases	Kafka to case record
Audit pipeline lag	<= 5 minutes	Time from decision to immutable record

Service level objectives.

Tier	RTO	RPO
Tier 1: scoring path (Gateway, Orchestrator, Feature Service, Online Store, FraudShield, Rules, Decision)	30 minutes	5 minutes
Tier 2: Case Management, Model Registry, Monitoring, Audit	4 hours	15 minutes
Tier 3: Offline store, Label service, analytics	24 hours	24 hours

Recovery objectives by tier.

3. Alert catalog

Alerts are defined in the observability repository. Each alert lists its severity and the procedure to follow. Alerts marked as added after an incident exist because that incident exposed a detection gap.

Alert ID	Name	Condition	Severity	Procedure
PRFIP-LAT-001	Scoring API latency elevated	p99 of /v1/risk/score above 100 ms for 5 minutes	Sev3	RB-06
PRFIP-LAT-002	Scoring API latency SLO breach	p99 of /v1/risk/score above 120 ms for 5 minutes	Sev2	RB-06
PRFIP-ERR-001	Scoring API error rate	HTTP 5xx above 0.1% for 5 minutes	Sev2	RB-06
PRFIP-TMO-001	Deadline exceeded rate	More than 0.5% of requests hit the 250 ms deadline for 5 minutes	Sev2	RB-06

Alert ID	Name	Condition	Severity	Procedure
PRFIP-TMO-002	Deadline exceeded rate critical	More than 2% of requests hit the 250 ms deadline for 5 minutes	Sev1	RB-06
PRFIP-DEG-001	Degraded decisions	More than 1% of decisions have degraded=true for 10 minutes	Sev2	RB-02
PRFIP-FS-AVAIL-001	Feature Service health check failing	3 consecutive failed health checks	Sev2	RB-02
PRFIP-FS-LAT-001	Feature Service latency	p99 above 25 ms for 5 minutes	Sev3	RB-02
PRFIP-FS-POOL-001	Redis connection pool utilization	Pool utilization above 75% for 5 minutes (added after INC-2026-041)	Sev3	RB-04
PRFIP-FS-FRESH-001	Feature freshness	p95 age of streaming features above 120 seconds for 5 minutes (added after INC-2026-057)	Sev2	RB-03
PRFIP-FSVC-001	FraudShield health check failing	3 consecutive failed health checks	Sev1	RB-01
PRFIP-INF-001	Model inference latency	FraudShield inference p99 above 40 ms for 10 minutes	Sev3	RB-01
PRFIP-INF-002	Model inference latency rollback criterion	Inference p99 above 60 ms for 15 minutes	Sev2	RB-05
PRFIP-KFK-001	Consumer lag	Stream Processor lag above 50,000 messages or 120 seconds	Sev2	RB-03
PRFIP-KFK-002	Dead-letter topic not empty	More than 10 messages in payments.transactions.dlq.v1	Sev2	RB-03
PRFIP-FLK-001	Flink checkpoint failures	2 consecutive checkpoint failures (routed to PagerDuty since 2026-09-09)	Sev2	RB-03
PRFIP-FLK-002	Flink restart loop	More than 3 job restarts in 15 minutes	Sev2	RB-03
PRFIP-RDS-001	Redis memory	Memory above 80% on any shard	Sev3	RB-04
PRFIP-RDS-002	Redis replica lag	Replica lag above 2 seconds for 5 minutes	Sev3	RB-04
PRFIP-DRF-001	Score drift warning	Score PSI above 0.20	Sev4	RB-08
PRFIP-DRF-002	Score drift critical	Score PSI above 0.25 for 2 consecutive hours	Sev2	RB-08
PRFIP-DRF-003	Feature drift	PSI above 0.25 on any monitored top-20 feature	Sev3	RB-08
PRFIP-PRC-001	Precision at decline threshold	Estimated precision below 65% for 24 hours	Sev2	RB-05
PRFIP-RVW-001	Manual review backlog	More than 2,500 open cases or oldest unassigned case above 45 minutes	Sev3	RB-09
PRFIP-DEC-001	Approval rate deviation	Approval rate deviates more than 3 percentage points from the 7-day baseline for 30 minutes	Sev3	RB-08
PRFIP-CASE-001	Case API errors	Case API 5xx above 1% for 10 minutes	Sev3	RB-09
PRFIP-SEC-001	Unauthorized snapshot access	Access to feature snapshot tables by a principal without the Confidential role	Sev1	Security incident process
PRFIP-AUD-001	Audit pipeline lag	Audit lag above 5 minutes	Sev3	RB-03
PRFIP-DR-001	DR replication lag	Replication lag to us-west-2 above 5 minutes	Sev3	RB-07

Alert ID	Name	Condition	Severity	Procedure
PRFIP-CERT-001	Certificate expiry	Any certificate expires in fewer than 14 days	Sev3	RB-10

Alert catalog.

4. Standard procedures

4.1 RB-01: FraudShield stops responding or fails health checks

Attribute	Value
Symptoms	PRFIP-FSVC-001 fires; requests show TIMEOUT_RULES_ONLY or MODEL_UNAVAILABLE; risk_score is null; degraded rate rises.
Impact	Decisions use rules only (Level 3). Fraud exposure rises with every minute; SEV-1.

Steps

1. Declare SEV-1 and page the model owner on-call. Post in the incident channel.
2. Confirm scope: check the degraded decision rate and decision_source split on the Scoring dashboard.
3. Check pod status: kubectl -n scoring get pods -l app=fraudshield; note CrashLoopBackOff or NotReady.
4. Check the last deployment: kubectl -n scoring rollout history deploy/fraudshield. If a release occurred within 2 hours, treat the release as the suspect.
5. Check the model artifact checksum in the pod log. A checksum mismatch means the registry entry and artifact differ; do not bypass the check.
6. If the fault follows a release, roll back the Scoring Service to the last known good release: kubectl -n scoring rollout undo deploy/fraudshield. Do not roll back to FraudShield v3.1; it is archived.
7. If pods are healthy but requests time out, check node saturation and inference latency (PRFIP-INF-001). Scale out: kubectl -n scoring scale deploy/fraudshield --replicas=64.
8. Verify recovery: degraded rate below 1%, p99 inference below 40 ms, decision_source MODEL above 99%.
9. Notify Fraud Operations of the duration of rules-only decisions so they can review the affected window.

Escalation: Escalate to the SRE Lead if not restored within 15 minutes.

4.2 RB-02: Feature Service unavailable or degraded

Attribute	Value
Symptoms	PRFIP-FS-AVAIL-001, PRFIP-FS-LAT-001 or PRFIP-DEG-001; decisions show decision_source MODEL_REDUCED_FEATURES and reason FEATURES_UNAVAILABLE.
Impact	Level 2 degradation: reduced-feature model v4.2-RF with stricter thresholds; manual review volume rises.

Steps

1. Declare SEV-2 (SEV-1 if degraded rate exceeds 25% for 15 minutes).
2. Check Feature Service pods: kubectl -n features get pods -l app=feature-service.
3. Check Redis health on the Feature Store dashboard: primary and replica status, memory and connection counts.

4 Check connection pool utilization (PRFIP-FS-POOL-001). At or near 100%, follow RB-04.

5 Check the last Feature Service or Scoring Service release; the number of lookups per request changed in SS-2.14.0 and caused INC-2026-041.

6 If the vendor circuit breaker is open (DeviceTrust), device features are null; this is Level 1 and needs no action beyond monitoring.

7 Restart unhealthy Feature Service pods one at a time; never restart all pods at once.

8 Inform Fraud Operations that manual review volume will be elevated; agree on temporary staffing.

9 Verify recovery: degraded decisions below 1%, Feature Service p99 below 25 ms.

Escalation: Escalate to Feature Platform Lead (Priya Raman) at once; to the SRE Lead if not recovered in 30 minutes.

4.3 RB-03: Kafka lag, Flink failure or stale features

Attribute	Value
Symptoms	PRFIP-KFK-001, PRFIP-KFK-002, PRFIP-FLK-001, PRFIP-FLK-002 or PRFIP-FS-FRESH-001; feature drift alert on velocity features.
Impact	Streaming features stale; the online store keeps last values for up to 6 hours; model inputs drift and fraud may be missed.

Steps

1 Declare SEV-2. Page the streaming owner (Wei Zhang or delegate).

2 Check the Flink job state in the Flink UI: RUNNING, RESTARTING or FAILED; note the exception.

3 If the exception is a deserialization error, check the DLQ topic and Schema Registry for a recent schema change on payments.transactions.v1.

4 Contact the Payment Gateway team (Naomi Fischer) and ask them to revert the producer if a breaking change was released.

5 Restart the job from the last good checkpoint: flink savepoint list, then restart with the chosen checkpoint.

6 After recovery, replay from the recorded offset and confirm 24-hour windows are recomputed.

7 Verify feature freshness p95 below 60 seconds and consumer lag near zero.

8 Notify Fraud Operations and the model owner of the stale-feature window so that decisions in that window can be reviewed.

Escalation: Escalate to the SRE Lead if stale for more than 60 minutes.

4.4 RB-04: Redis saturation or connection pool exhaustion

Attribute	Value
Symptoms	PRFIP-FS-POOL-001, PRFIP-RDS-001, PRFIP-RDS-002, PRFIP-FS-LAT-001; Feature Service latency high with low Redis CPU.
Impact	Feature Service exceeds its 80 ms budget; requests fall to Level 2; end-to-end latency rises. This is the failure mode of INC-2026-041.

Steps

1. Declare SEV-2.
2. Check pool utilization per pod: the redis_pool_in_use metric against the configured limit of 480.
3. Check the lookups per request metric. A rise after a release points to a call pattern change; roll back that release.
4. Check Redis cluster CPU, memory and shard balance. A hot shard indicates key skew; a failed primary indicates failover (about 30 seconds).
5. Scale Feature Service pods only if pool limits allow; adding pods multiplies total connections and can worsen the problem.
6. If pool exhaustion follows a release of the Scoring Service, roll back: kubectl -n scoring rollout undo deploy/scoring-service.
7. Verify recovery: pool utilization below 60%, Feature Service p99 below 25 ms, deadline exceeded rate below 0.5%.

Escalation: Escalate to the Feature Platform Lead (Priya Raman).

4.5 RB-05: Model rollback procedure

Attribute	Value
Symptoms	One of the rollback conditions in Part 3 holds: score PSI above 0.25 for two hours with lower precision, precision at decline below 65% for 24 hours, inference p99 above 60 ms for 15 minutes, or approval rate deviation above 3 points for 30 minutes.
Impact	Model quality or latency has degraded; the decision needs both engineering and model-owner authorization.

Steps

1. Confirm that the condition holds using the Model Monitoring dashboard; record screenshots and times.
2. Obtain authorization from the Incident Commander and the model owner (Sarah Johnson) or delegate; record both approvals in the incident channel.
3. Identify the rollback target: the previous approved serving configuration of FraudShield v4.2. FraudShield v3.1 is archived and must not be used.
4. If the fault is in the model artifact itself and no earlier approved configuration is valid, enable v4.2-RF (Level 2) or rules-only (Level 3) with Head of Fraud Engineering approval.
5. Raise an emergency change (CHG) and run: kubectl -n scoring rollout undo deploy/fraudshield.
6. Confirm the models.events.v1 rollback event and audit entry were emitted.
7. Verify: PSI, precision proxy, approval rate and latency return within thresholds within 60 minutes.
8. Open a model incident review with Model Risk & Validation within 2 business days.

Escalation: The model owner and Head of Model Risk are informed of every rollback.

4.6 RB-06: Elevated timeouts or latency on /v1/risk/score

Attribute	Value
Symptoms	PRFIP-LAT-001, PRFIP-LAT-002, PRFIP-ERR-001, PRFIP-TMO-001 or PRFIP-TMO-002.
Impact	Requests exceed the 250 ms deadline and receive rules-only decisions with reason TIMEOUT_RULES_ONLY.

Steps

1. Declare SEV-2 (SEV-1 for PRFIP-TMO-002).
2. Read the latency budget breakdown on the Scoring dashboard to find the slow step: Feature Service, vendor adapter, Merchant Risk, FraudShield, Rules.
3. Follow the procedure for the slow component: RB-02 or RB-04 for Feature Service, RB-01 for FraudShield.
4. Check for a release in the last 2 hours across the scoring path; roll back if correlated.
5. Check traffic: a spike above 6,000 requests per second requires scaling out the Orchestrator: kubectl -n scoring scale deploy/orchestrator --replicas=120.
6. Check API Gateway health and rate-limit rejections.
7. Verify recovery: p99 below 120 ms and deadline exceeded rate below 0.5%.

Escalation: Escalate to the SRE Lead if not resolved in 30 minutes.

4.7 RB-07: Regional failover to us-west-2

Attribute	Value
Symptoms	Loss of us-east-1 or a decision by the Head of Fraud Engineering to fail over.
Impact	Scoring unavailable in the primary region; target RTO 30 minutes, RPO 5 minutes.

Steps

1. Declare SEV-1; the Head of Fraud Engineering authorizes failover.
2. Confirm the outage is regional and not a component fault (check the external status page and cross-region synthetic checks).
3. Notify the Payment Gateway team that traffic will switch; agree the moment of DNS change.
4. Scale the DR environment from 50% to 100% capacity using the DR scaling job.
5. Promote the Aurora cross-region replica and confirm the Redis DR cluster contains replicated data.
6. Replay Kafka from the mirrored offsets to rebuild the last minutes of features.
7. Switch traffic by updating the weighted DNS record for the API endpoint; confirm requests in the DR region.
8. Verify health: SLOs on the DR dashboard, decision_source MODEL above 99%.
9. Keep incident open until the primary region is restored; plan failback in a change window.

Escalation: Failback requires an approved change.

4.8 RB-08: Drift, approval-rate or stale merchant score alerts

Attribute	Value
Symptoms	PRFIP-DRF-001, PRFIP-DRF-002, PRFIP-DRF-003, PRFIP-DEC-001.
Impact	Data or behavior has changed; may indicate a data pipeline fault, a real fraud attack or a model issue.

Steps

1. Identify which feature or score is drifting and when the drift began.

2 Check for a recent upstream schema, pipeline or release change; compare against the Change calendar.
3 Check feature freshness: stale features are the most common cause of feature drift (see INC-2026-057).
4 Compare drift by segment (channel, region, card type).
5 If a data fault, follow RB-03. If a possible attack, notify Fraud Operations and the model owner.
6 If PRFIP-DRF-002 or PRFIP-PRC-001 persists, start RB-05.

Escalation: Model owner decides on retraining or threshold review through the Model Governance Committee.

4.9 RB-09: Manual review backlog and Case Management issues

Attribute	Value
Symptoms	PRFIP-RVW-001 or PRFIP-CASE-001.
Impact	Review decisions wait longer; customers experience delays.

Steps

1 Check the review queue size and oldest case age in the Case dashboard.
2 Check whether degraded decisions are inflating the queue (PRFIP-DEG-001).
3 Ask the Fraud Operations Manager (Tom Alvarez) to add analysts or rebalance queues.
4 If Case Management is down, confirm REVIEW events are queuing in fraud.cases and risk.decisions topics and export the emergency queue.
5 After recovery, verify that queued cases are created and that timestamps are preserved.

Escalation: Fraud Operations decides on temporary threshold adjustments; these need model owner and Head of Model Risk approval.

4.10 RB-10: Deployment, canary and certificate procedures

Attribute	Value
Symptoms	Planned releases and certificate expiry alert PRFIP-CERT-001.
Impact	Releases carry risk; every Tier 1 release follows the checklist.

Steps

1 Confirm an approved CHG and the Release Readiness Review outcome.
2 Confirm the load test at 120% of the trailing 30-day peak using the release's actual call pattern (checklist item RC-11).
3 Deploy during the change window: Tuesday 10:00 to 12:00 UTC for Tier 1 services (excluding freeze periods).
4 Start the canary at 5% and keep it for at least 24 hours, including a peak traffic period.
5 Watch latency, deadline exceeded rate, Feature Service pool utilization and degraded rate; abort if any exceeds its threshold.
6 Promote to 25%, 50% and 100% with the same checks.

7 For expiring certificates, rotate through the certificate authority workflow; verify with the synthetic check.

Escalation: Abort and roll back on any threshold breach.

5. Diagnostic commands and queries

```
# Degraded decision share (last 10 minutes)
sum(rate(risk_decisions_total{degraded="true"}[10m])) / sum(rate(risk_decisions_total[10m]))

# End-to-end p99 latency
histogram_quantile(0.99, sum(rate(score_latency_seconds_bucket[5m])) by (le))

# Feature Service pool utilization
max(redis_pool_in_use / redis_pool_limit) by (pod)

# Feature freshness p95
histogram_quantile(0.95, sum(rate(feature_age_seconds_bucket[5m])) by (le))

# Kafka consumer lag for the stream processor
kafka_consumergroup_lag{group="prfip-stream-processor"}
```

6. Change windows and freezes

Tier 1 releases are deployed in the Tuesday 10:00 to 12:00 UTC window. Emergency changes require the approval of the Incident Commander. Change freezes apply to the last two weeks of the fiscal year and to the week of major shopping events announced by the business. Every change needs a rollback plan, and every Tier 1 change needs the load test evidence described in RB-10.

7. Post-incident actions

1 Publish a timeline within 24 hours of resolution.

2 Hold a blameless review within 5 business days.

3 Publish the postmortem within 10 business days for SEV-1 and SEV-2 incidents.

4 Track every action item with an owner and due date; report open items weekly to the Head of Fraud Engineering.

5 Update this runbook, alerts and checklists in the same change that closes the action item.

Part 6. Incident and Postmortem Reports

1. Incident log

Incident	Date	Severity	Title	Root cause	Status
INC-2026-011	2026-02-19	SEV-3	Merchant Risk Service daily job late	Airflow worker out of memory	Closed
INC-2026-019	2026-03-27	SEV-3	Device Intelligence Adapter vendor timeouts	Vendor regional outage; circuit breaker opened as designed	Closed
INC-2026-027	2026-05-05	SEV-2	Case creation delay of 40 minutes	Kafka consumer group rebalance in Case Management	Closed
INC-2026-033	2026-06-16	SEV-3	Elevated timeouts after certificate rotation	Expired intermediate certificate on one Orchestrator pool	Closed
INC-2026-041	2026-08-11	SEV-2	Fraud scoring latency degradation	Redis connection pool exhaustion after SS-2.14.0	Closed (actions open: 1)
INC-2026-048	2026-08-24	SEV-4	Dashboard data gap	Prometheus remote-write lag	Closed
INC-2026-057	2026-09-03	SEV-2	Feature pipeline failure	Breaking schema change without compatibility enforcement	Closed (actions open: 2)

Incident log, February to September 2026.

2. INC-2026-041: August 2026 Fraud Platform Postmortem

Attribute	Value
Incident	INC-2026-041 (also published as the August 2026 Fraud Platform Postmortem)
Title	Fraud scoring latency degradation
Date	2026-08-11 (Tuesday)
Severity	SEV-2
Impact window	13:52 to 15:41 UTC (1 hour 49 minutes)
Incident commander	Daniel Okafor
Primary on-call	Ravi Menon
Postmortem author	Daniel Okafor with Priya Raman
Postmortem published	2026-08-25
Status	Closed; one action item open

2.1 Summary

On 11 August 2026 the p99 latency of the Fraud Scoring API rose from about 84 ms to over 400 ms during the lunch-hour traffic peak, and a share of requests exceeded the 250 ms deadline. The cause was the release of Scoring Service SS-2.14.0 one week earlier, which increased the number of Feature Service lookups per scoring request from 6 to 14. The Feature Service Redis client connection pool was sized at 200 connections per pod for the earlier call pattern. At the peak of 4,800 requests per second the pool was exhausted, requests queued inside the Feature Service, and lookups exceeded their 80 ms budget. The Orchestrator responded by falling back to the reduced-feature model for many requests and to rules-only decisions when the 250 ms deadline was reached. Service was restored by rolling back to SS-2.13.2.

2.2 Impact

Measure	Value
Duration of customer impact	1 hour 49 minutes (13:52 to 15:41 UTC)
Peak traffic	4,800 requests per second (typical peak 3,900)
Peak Scoring API p99	410 ms (baseline 84 ms)
Feature Service p99 at peak	610 ms (SLO 25 ms)
Requests decided by the reduced-feature model	About 22% during the window
Requests exceeding the 250 ms deadline (rules-only)	4.1%, roughly 1.1 million requests
Manual review queue	Up 38% (about 700 additional cases)
Estimated additional fraud loss	USD 186,000
Estimated declined legitimate volume	USD 1.9 million gross payment volume
Customer-facing errors	None; all requests received a decision
SLO effect	August latency SLO compliance 97.2% against a 99% target

Impact of INC-2026-041.

2.3 Timeline (UTC, 2026-08-11)

Time	Event
13:40	Traffic ramps to lunch-hour peak; 4,800 requests per second against a typical 3,900.
13:52	Alert PRFIP-LAT-002 fires: /v1/risk/score p99 above 120 ms. On-call engineer Ravi Menon acknowledges.
14:07	Ravi Menon observes Feature Service p99 at 610 ms and escalates to Daniel Okafor.
14:14	Incident declared SEV-2. Daniel Okafor becomes incident commander; incident channel opened.
14:31	Fraud Operations (Tom Alvarez) notified: manual review queue growing (up 38%).
14:48	Team suspects a Redis shard problem and fails over shard 7; no improvement.
15:12	Priya Raman's team identifies Redis connection pool exhaustion: 200 of 200 connections in use per Feature Service pod.
15:26	Decision to roll back Scoring Service SS-2.14.0 to SS-2.13.2, which returns to 6 grouped lookups per request instead of 14.
15:33	Rollback complete on all pods; latency begins to fall.
15:41	Scoring API p99 returns to 88 ms; deadline-exceeded rate returns to baseline.
16:02	Incident moved to monitoring; declared resolved at 17:30 after two stable hours.

Timeline of INC-2026-041.

2.4 Technical analysis

Until SS-2.14.0 the Scoring Service issued six grouped Feature Service calls per request, one per entity family, with a Redis connection pool of 200 per Feature Service pod. SS-2.14.0 was released on 2026-08-04 under CHG-2026-0388 to prepare for the FraudShield v5.0 feature families (ADR-011). It changed the client to issue 14 per-family calls, which increased the demand for pooled connections by more than a factor of two. The pool limit was not reviewed as part of the release. Under normal load, utilization stayed below the limit and latency remained within target, so the change was not noticed in the first week.

At 13:40 UTC traffic ramped toward the daily lunch-hour peak. As concurrent lookups rose, connections in each pod were all in use (200 of 200), so lookups waited for a free connection. Redis itself was healthy: CPU stayed under 35% and no shard failed. The wait time appeared as Feature Service latency, which reached 610 ms. The Orchestrator's 80 ms Feature Service budget was exceeded for a large share of requests, moving them to the reduced-feature model, and where the wait combined with other delays the 250 ms deadline was reached and a rules-only decision was returned. Because Feature Service threads were busy waiting, latency for all requests rose.

2.5 Root cause and contributing factors

Root cause: Redis connection pool exhaustion in the Feature Service, caused by the SS-2.14.0 change from 6 to 14 lookups per request without a corresponding increase of the connection pool (200 per pod) or batching of lookups.

- **Load test did not represent production.** The staging load test ran at 40% of production peak and used the earlier call pattern of 6 lookups per request.
- **Canary did not include a peak.** The canary of 5% for 2 hours ran during a low-traffic period and did not reach the concurrency at which the pool saturates.

- **No alert on pool utilization.** Connection pool utilization was not monitored; the first alert was the customer-facing latency alert PRFIP-LAT-002.
- **Detection path.** The first hypothesis (a Redis shard fault) cost about 40 minutes; failing over shard 7 at 14:48 had no effect.
- **Release review accepted a known risk.** The Release Readiness Review RRR-2026-031 noted the 40% load test profile and accepted it with a condition to monitor pool metrics, but no monitoring existed for pool utilization.

2.6 What went well

- The degradation ladder worked: every request received a decision and there were no customer-facing errors.
- The alert fired within 12 minutes of first impact.
- The rollback was quick once the cause was identified (7 minutes from decision to completion).
- Fraud Operations was notified early and adjusted staffing.

2.7 What went poorly

- The pool exhaustion was identified 80 minutes after the alert.
- No dashboard showed connection pool utilization.
- Rules-only decisions and reduced-feature decisions increased fraud exposure and false declines.
- The staging environment could not reproduce production concurrency.

2.8 Corrective actions

ID	Action	Owner	Due	Status
AI-041-1	Raise Redis connection pool from 200 to 480 per Feature Service pod	Priya Raman	2026-08-14	Done (SS-2.14.1, 2026-08-25)
AI-041-2	Replace per-family calls with pipelined MGET, reducing to 3 round trips	Priya Raman	2026-08-25	Done (SS-2.14.1)
AI-041-3	Add alert PRFIP-FS-POOL-001 for pool utilization above 75%	Daniel Okafor	2026-08-18	Done
AI-041-4	Update load test profile to 120% of trailing 30-day peak using current call pattern	Daniel Okafor	2026-08-28	Done
AI-041-5	Add release checklist item RC-11 requiring a load test at 120% of peak before Tier 1 releases	Sarah Johnson	2026-08-31	Done
AI-041-6	Require canary window to include a peak traffic period	Marcus Lee	2026-09-04	Done
AI-041-7	Amend ADR-011 to require batching and pool sizing review	Marcus Lee	2026-08-27	Done
AI-041-8	Review Fraud Operations staffing plan for degraded-mode surges	Tom Alvarez	2026-10-15	Open

Action items for INC-2026-041.

2.9 Permanent fix

Scoring Service SS-2.14.1 was released on 2026-08-25 (REL-2026-033, CHG-2026-0411). It replaces 14 per-family calls with pipelined MGET commands that use 3 Redis round trips per request, and raises the Feature Service connection pool from 200 to 480 per pod. New alert PRFIP-FS-POOL-001 fires at 75% pool utilization. The release checklist gained item RC-11 (load test at 120% of trailing 30-day peak with the actual call pattern), and canary windows must include a peak traffic period. ADR-011 was amended on 2026-08-27 to require batching and pool sizing review.

2.10 Lessons learned

- A change in call pattern is a capacity change; it must be load tested and its resource limits reviewed.
- Alerts should exist for saturation of every shared resource on the scoring path.
- Canary evidence is only valid if the canary sees representative load.

3. INC-2026-057: Feature pipeline failure

Attribute	Value
Incident	INC-2026-057
Title	Feature pipeline failure: stale streaming features
Date	2026-09-03 (Thursday)
Severity	SEV-2
Impact window	08:41 to 11:53 UTC (3 hours 12 minutes)
Incident commander	Wei Zhang
Detected by	Drift alert PRFIP-DRF-003 at 09:20 UTC (39 minutes after start)
Status	Closed; two action items open

3.1 Summary

On 3 September 2026 the Payment Gateway team released a producer change that altered the type of the amount field in payments.transactions.v1 from integer minor units to a decimal string, registered as schema version 1.7. The subject in Schema Registry was configured with compatibility NONE for historical reasons, so the registration succeeded although consumers could not read the new format. The Flink stream processor failed to deserialize the new records and entered a restart loop. No streaming features were updated for 3 hours and 12 minutes. The online store kept serving the last values because streaming features have a TTL of 6 hours, so scoring continued without errors but with stale behavioral features such as device_velocity_24h.

3.2 Impact

Measure	Value
Duration of stale features	3 hours 12 minutes (08:41 to 11:53 UTC), plus 47 minutes of recovery replay
Features affected	All 33 streaming behavioral features (velocity, counts, sums)
Decisions made with stale features	About 39 million
Score PSI at peak	0.19; PSI on device_velocity_24h reached 0.31
Estimated missed fraud	USD 94,300 (transactions that would likely have been flagged by velocity features)
Customer-facing errors	None
Fraud Operations effect	Fewer velocity-related review flags

Impact of INC-2026-057.

3.3 Timeline (UTC, 2026-09-03)

Time	Event
08:38	Payment Gateway team deploys producer release changing the amount field in payments.transactions.v1 from integer minor units to a decimal string, under schema version 1.7.
08:41	Flink job fails to deserialize new records and starts restarting repeatedly; streaming feature updates stop.
08:47	Checkpoint failure alert posts to the low-priority Slack channel #prfip-info; nobody is paged.
09:20	Drift monitor raises PSI 0.31 on device_velocity_24h (PRFIP-DRF-003). On-call engineer investigates.
09:52	Fraud Operations reports fewer velocity-related flags in the review queue.
10:15	Wei Zhang confirms the Flink restart loop and identifies the schema change.
10:40	Incident declared SEV-2; commander Wei Zhang. Payment Gateway team asked to revert the producer.
11:38	Producer reverted to schema 1.6.
11:53	Flink job restarted from the last good checkpoint; stream processing resumes.
12:40	Backlog replayed and 24-hour windows recomputed; freshness back within target.
13:10	Incident resolved.

Timeline of INC-2026-057.

3.4 Root cause and contributing factors

Root cause: A breaking schema change to payments.transactions.v1 was registered and deployed without compatibility enforcement in Schema Registry, and the stream processor had no dead-letter handling for records it could not deserialize.

- **Compatibility was NONE.** The subject had been set to NONE during a 2024 migration and never restored.
- **No dead-letter path.** A single undeserializable record caused the job to fail and restart from the last checkpoint indefinitely.
- **Alert routing.** The Flink checkpoint failure alert (PRFIP-FLK-001) posted to a low-priority Slack channel and paged no one.

- **Detection through model monitoring, not pipeline monitoring.** The first page came from feature drift on device_velocity_24h, 39 minutes after the start.
- **No cross-team notification.** The Payment Gateway team did not know the stream processor consumed the amount field as an integer.
- **TTL of 6 hours** hides stale features from scoring errors; staleness is invisible without a freshness metric.

3.5 Recovery

The producer was reverted to schema 1.6 at 11:38 UTC. The Flink job was restarted from its last good checkpoint at 11:53 and the Kafka backlog was replayed until 12:40, when 24-hour windows had been recomputed and freshness was back within the 60-second target. Fraud Operations reviewed the flagged window and confirmed the estimate of missed fraud.

3.6 Corrective actions

ID	Action	Owner	Due	Status
AI-057-1	Enforce BACKWARD_TRANSITIVE compatibility on all PRFIP subjects	Wei Zhang	2026-09-09	Done (REL-2026-036)
AI-057-2	Add dead-letter routing for deserialization failures in the Flink job	Wei Zhang	2026-09-09	Done
AI-057-3	Route Flink checkpoint failure alerts to PagerDuty (PRFIP-FLK-001)	Daniel Okafor	2026-09-07	Done
AI-057-4	Add feature freshness alert PRFIP-FS-FRESH-001 (p95 above 120 s)	Priya Raman	2026-09-11	Done
AI-057-5	Cross-team change notification for schema changes affecting payments.transactions.v1	Naomi Fischer	2026-09-18	Done
AI-057-6	Add contract tests between Payment Gateway producer and Flink consumer to CI	Naomi Fischer	2026-10-30	Open
AI-057-7	Review TTL of streaming features in online store (currently 6 hours)	Priya Raman	2026-10-15	Open

Action items for INC-2026-057.

3.7 Permanent fix

REL-2026-036 (CHG-2026-0421) on 2026-09-09 enforced BACKWARD_TRANSITIVE compatibility on every PRFIP subject (ADR-012), added dead-letter routing so that undeserializable records go to payments.transactions.dlq.v1 without stopping the job, routed alert PRFIP-FLK-001 to PagerDuty, and added the feature freshness alert PRFIP-FS-FRESH-001 (p95 above 120 seconds).

4. Patterns across incidents

Theme	INC-2026-041	INC-2026-057	Response
Change introduced the failure	Scoring Service call pattern change	Producer schema change	Contract tests, load tests at representative scale, cross-team notification
Detection gap	No pool utilization alert	Pipeline alert not paging; no freshness metric	New alerts PRFIP-FS-POOL-001 and PRFIP-FS-FRESH-001
Graceful degradation	Worked (Level 2 and 3)	Not applicable; stale not degraded	Freshness now treated as a degradation trigger candidate
Time to identify cause	80 minutes	95 minutes from first alert to confirmed cause	Runbook procedures RB-04 and RB-03

Comparison of the two incidents.

Related documents: System Architecture Design (failure scenarios FS-4 and FS-6, ADR-011 and ADR-012), Production Runbook (RB-03, RB-04, RB-05), Fraud Model Technical Design (feature pipeline and monitoring), Release Checklist (RC-11).

Part 7. Disaster Recovery Architecture

Attribute	Value
Document	Disaster Recovery Architecture
Version / status	v2.2 / CURRENT (Approved)
Effective date	2026-07-20
Owner	Marcus Lee
Related procedure	Runbook RB-07

Document control for Disaster Recovery Architecture.

1. Strategy

The platform runs in a single active region (us-east-1) with a warm standby in us-west-2 (ADR-007). Active-active operation was proposed in ADR-010 and has not been approved. The standby holds replicated data and runs at 50% of production capacity; on failover it scales to 100%. The scoring path is designed to recover within 30 minutes with at most 5 minutes of lost data.

Tier	Components	RTO	RPO	Recovery method
1	Scoring path: API Gateway, Orchestrator, Feature Service, Online store, FraudShield, Rules, Decision	30 minutes	5 minutes	Scale DR cluster, promote replicas, replay Kafka, switch DNS
2	Case Management, Model Registry, Monitoring, Audit	4 hours	15 minutes	Restore from cross-region replicas and backups
3	Offline store, Label service, analytics	24 hours	24 hours	Rebuild from S3 replication

Recovery tiers.

2. Replication and backup

Data	Method	Lag target	Alert
Online feature store (Redis)	Global datastore replication	under 5 minutes	PRFIP-DR-001
Kafka	MirrorMaker 2 to us-west-2	under 2 minutes	PRFIP-DR-001
Aurora (cases, MLflow)	Cross-region read replica	under 1 minute	Replica lag alert
S3 data lake and models	Cross-region replication	under 15 minutes	Replication metrics
Audit store	S3 Object Lock plus Azure immutable copy	under 5 minutes	PRFIP-AUD-001
Configuration	Git; Terraform state replicated	on commit	CI check

Replication and backup.

3. Failover test history

Test	Date	Result	Failover time	Data loss	Findings
DR-2026-Q1	2026-03-18	Passed	31 minutes	4.1 minutes	DNS TTL of 300 seconds slowed switch; reduced to 60 seconds
DR-2026-Q2	2026-06-17	Passed	27 minutes	3.2 minutes	Aurora replica promotion runbook step missing; added
DR-2026-Q3	2026-09-16	Passed	24 minutes	2.8 minutes	No blocking findings; scaling job optimized

Quarterly failover test results.

4. Failure of the disaster recovery region

If the standby region is unavailable, the platform continues in the primary region and alert PRFIP-DR-001 raises a SEV-3 incident, because the RPO objective is at risk. A second failure in the primary region during this period would be a SEV-1 with recovery from backups only. Recovery from backups has a longer RTO of 8 hours and is exercised annually.

Part 8. Technical Appendices

The appendices provide reference detail for the parts above: message schemas, configuration, telemetry, validation evidence, additional runbook procedures, the threat model and earlier incident summaries. Each appendix carries the version and status of the part it supports.

1. Appendix A: Message schemas and configuration reference

1.1 Schema of payments.transactions.v1 (schema version 1.6 current)

The transaction event is produced by the Payment Gateway and consumed by the stream processor and the data lake sink. Schema version 1.6 is the current version; version 1.7, which changed amount to a decimal string, was reverted on 2026-09-03 after INC-2026-057 and was never a valid version. Fields, types and classification are defined in the Transaction Data Specification in the business documentation set.

Field	Meaning	Type	Source	Nullable	Class
transaction_id	Unique identifier of the payment transaction	string (UUID)	Payment Gateway	No	Confidential
request_id	Unique identifier of the scoring request; used for tracing	string (UUID)	API Gateway	No	Confidential
idempotency_key	Client-generated key preventing duplicate scoring	string	Payment Gateway	No	Internal
event_time	Time the transaction was initiated (UTC)	timestamp	Payment Gateway	No	Confidential
received_time	Time the platform received the request (UTC)	timestamp	API Gateway	No	Internal
merchant_id	Merchant identifier	string	Merchant Platform	No	Internal
merchant_country	ISO 3166-1 alpha-2 country of the merchant	string(2)	Merchant Platform	No	Internal
mcc	Merchant category code	string(4)	Merchant Platform	No	Internal
card_token_id	Network or vault token that stands in for the card number; the PAN is never received by the platform	string	Token Vault (Payment Gateway)	No	Confidential
card_bin	First 8 digits of the card number	string(8)	Payment Gateway	No	Confidential
card_last4	Last four digits of the card number	string(4)	Payment Gateway	No	Confidential
card_brand	Card network (VISA, MASTERCARD, AMEX, DISCOVER)	string	Payment Gateway	No	Internal
customer_id	Merchant-side customer identifier	string	Payment Gateway	Yes	Confidential
customer_email_hash	SHA-256 hash of the lowercased customer email with platform salt	string(64)	Payment Gateway	Yes	Confidential

Field	Meaning	Type	Source	Nullable	Class
customer_phone_hash	SHA-256 hash of the normalized phone number	string(64)	Payment Gateway	Yes	Confidential
billing_country	ISO country of the billing address	string(2)	Payment Gateway	Yes	Confidential
billing_postal_code	Postal code of the billing address	string	Payment Gateway	Yes	Confidential
shipping_country	ISO country of the shipping address	string(2)	Payment Gateway	Yes	Confidential
shipping_postal_code	Postal code of the shipping address	string	Payment Gateway	Yes	Confidential
ip_address	Client IP address of the purchaser	string (IPv4/IPv6)	Payment Gateway	Yes	Confidential
ip_country	Country derived from IP geolocation	string(2)	Device Intelligence Adapter	Yes	Confidential
device_id	Platform-assigned device identifier	string	Payment Gateway	Yes	Confidential
device_fingerprint_hash	Hash of browser or device attributes	string(64)	Payment Gateway	Yes	Confidential
user_agent	HTTP user agent string	string	Payment Gateway	Yes	Internal
session_id	Merchant session identifier	string	Payment Gateway	Yes	Confidential
amount_minor_units	Transaction amount in minor currency units (for example cents)	integer (int64)	Payment Gateway	No	Confidential
currency_code	ISO 4217 currency code	string(3)	Payment Gateway	No	Internal
amount_usd	Amount converted to USD	decimal(12, 2)	Payment Gateway	No	Confidential
payment_method_type	Payment instrument type	string	Payment Gateway	No	Internal
entry_mode	Capture mode of the payment	string	Payment Gateway	No	Internal
is_recurring	True for merchant-initiated recurring payments	boolean	Payment Gateway	No	Internal
three_ds_status	3-D Secure outcome (AUTHENTICATED, ATTEMPTED, FAILED, NOT_ENROLLED, NOT_APPLICABLE)	string	Payment Gateway	No	Internal
three_ds_version	3-D Secure protocol version	string	Payment Gateway	Yes	Internal
avs_result	Address verification result	string(1)	Payment Gateway	Yes	Internal
cvv_result	Card verification result	string(1)	Payment Gateway	Yes	Internal

Field	Meaning	Type	Source	Nullable	Class
auth_response_code	Issuer authorization response code	string(2)	Payment Gateway	Yes	Internal
network_transaction_id	Card network transaction reference	string	Payment Gateway	Yes	Confidential
acquirer_id	Acquirer identifier	string	Payment Gateway	No	Internal
order_id	Merchant order identifier	string	Payment Gateway	Yes	Confidential
order_item_count	Number of order line items	integer	Payment Gateway	Yes	Internal
order_has_digital_goods	True when the order includes digital goods	boolean	Payment Gateway	Yes	Internal
shipping_method	Selected shipping method	string	Payment Gateway	Yes	Internal
channel	Sales channel (WEB, MOBILE_APP, IN_STORE, CALL_CENTER)	string	Payment Gateway	No	Internal
api_client_id	OAuth client that submitted the request	string	API Gateway	No	Internal

Fields of the transaction event.

1.2 Schema of scoring outputs

Field	Meaning	Type	Source	Nullable	Classification
risk_score	Calibrated fraud probability between 0 and 1; null when decision_source is RULES_ONLY	decimal(5,4)	FraudShield Scoring Service	Yes	Confidential
decision	Final decision: APPROVE, STEP_UP, REVIEW or DECLINE	string	Decision Service	No	Confidential
decision_id	Unique identifier of the decision for later lookup	string (UUID)	Risk Orchestrator	No	Confidential
decision_reason_codes	List of reason codes explaining the decision	array of string	Decision Service	No	Confidential
model_version	FraudShield version that produced the score (for example v4.2 or v4.2-RF)	string	FraudShield Scoring Service	Yes	Internal
threshold_set_version	Version of the threshold configuration applied	string	Decision Service	No	Internal
rule_set_version	Version of the rule set applied	string	Rules Engine	No	Internal
decision_source	MODEL, MODEL_REDUCED_FEATURES or RULES_ONLY	string	Risk Orchestrator	No	Internal
degraded	True when the decision was produced under any degradation mode	boolean	Risk Orchestrator	No	Internal
latency_ms	Server-side processing time in milliseconds	integer	Risk Orchestrator	No	Internal
feature_snapshot_ref	Reference to the stored feature snapshot used for this decision	string	Risk Orchestrator	No	Internal
case_id	Case identifier when decision is REVIEW	string	Case Management Service	Yes	Confidential

Fields of risk.scores.v1 and risk.decisions.v1.

1.3 Decision reason codes

Reason code	Meaning
HIGH_DEVICE_VELOCITY	device_velocity_1h or device_velocity_24h far above the device's normal level
HIGH_CARD_VELOCITY	Unusual number of transactions by the card in a short period
NEW_DEVICE	Device first seen recently combined with elevated amount
MULTI_CARD_DEVICE	Many distinct cards used on one device
GEO_VELOCITY_IMPOSSIBLE	Implied travel speed between transactions is not plausible
IP_COUNTRY_MISMATCH	IP country differs from billing country
PROXY_OR_VPN	IP is a proxy, VPN or data center range
HIGH_RISK_MERCHANT	merchant_risk_score above the merchant risk threshold
AMOUNT_ANOMALY	Amount far above the card's average
EMULATOR_OR_ROOTED_DEVICE	Device flagged as emulator or rooted
CARD_TESTING_PATTERN	Pattern of small amounts followed by a larger amount
AVS_CVV_FAILURE	Address or CVV verification failed
DEVICE_SIGNAL_MISSING	Vendor device signals were unavailable
FEATURES_UNAVAILABLE	Feature Service unavailable; reduced-feature model used
RULE_HARD_DECLINE	Hard-decline rule matched
RULE_SANCTIONS_MATCH	Sanctions screening matched
TIMEOUT_RULES_ONLY	250 ms deadline exceeded; rules-only decision
STEP_UP_REQUIRED	3-D Secure step-up requested

Reason codes returned in decision_reason_codes.

1.4 Rules catalog

Rule	Name	Condition	Action	Owner	Expiry
R-001	Sanctions match	Payer or merchant matches sanctions list	DECLINE	Compliance	No expiry
R-002	Blocked BIN	Card BIN on blocked BIN list	DECLINE	Fraud Operations	2027-03-31
R-003	Blocked device	Device ID on confirmed-fraud device list	DECLINE	Fraud Operations	No expiry
R-004	Card velocity guard	More than 20 transactions by the card in 1 hour	DECLINE	Fraud Engineering	No expiry
R-005	Multi-card device guard	More than 8 distinct cards on one device in 1 hour	REVIEW	Fraud Operations	2026-12-31
R-006	High-value first purchase	Amount above USD 2,500 and customer_account_age_days below 2	REVIEW	Fraud Operations	2026-12-31
R-007	Card testing	5 or more declines in 10 minutes on the same device	DECLINE	Fraud Engineering	No expiry
R-008	Impossible travel	geo_velocity_kmh_card above 1,000	REVIEW	Fraud Operations	2027-06-30
R-009	Rules-only high amount	Only in degraded rules-only mode: amount above USD 2,000	REVIEW	Fraud Engineering	No expiry
R-010	Rules-only high-risk merchant	Only in rules-only mode: merchant_risk_score above 0.80	DECLINE	Fraud Engineering	No expiry
R-011	Trusted customer allow	Customer with 24 or more months history and no chargebacks, amount below USD 300	APPROVE	Fraud Operations	2027-01-31
R-012	Recurring low-risk allow	Recurring payment, same amount as previous, no failures in 90 days	APPROVE	Fraud Operations	2026-11-30

Active rules in rule set RS-2026.09.

1.5 Configuration parameters

Component	Parameter	Value	Changed
Risk Orchestrator	request.deadline.ms	250	2026-02-10 (ADR-009)
Risk Orchestrator	budget.feature_service.ms	80	2025-09-04
Risk Orchestrator	budget.device_adapter.ms	60	2025-09-04
Risk Orchestrator	budget.merchant_risk.ms	40	2025-09-04
Risk Orchestrator	budget.fraudshield.ms	40	2025-09-04
Risk Orchestrator	budget.rules.ms	15	2025-09-04
Risk Orchestrator	fallback.reduced_model	v4.2-RF	2026-04-06
Risk Orchestrator	degraded.thresholds.decline	0.92	2026-04-06
Risk Orchestrator	degraded.thresholds.review	0.75	2026-04-06
Risk Orchestrator	rules_only.review_amount_usd	2000	2026-02-10
Feature Service	redis.pool.max_connections	480	2026-08-25 (was 200)
Feature Service	redis.command_timeout.ms	30	2025-09-04
Feature Service	lookup.batching	pipelined MGET, 3 round trips	2026-08-25
Feature Service	feature.max_age.streaming.h	6	2025-09-04
Feature Service	feature.max_age.batch.h	36	2025-09-04
Scoring Service	model.production	fraudshield-v4.2	2026-04-06
Scoring Service	model.shadow	fraudshield-v5.0	2026-08-19
Scoring Service	model.fallback	fraudshield-v4.2-rf	2026-04-06
Scoring Service	threshold_set	TS-4.2.1	2026-04-06
Scoring Service	inference.warmup.requests	2000	2025-11-20
Device Adapter	cache.ttl.minutes	15	2025-09-04
Device Adapter	breaker.failures	5 consecutive	2025-09-04
Device Adapter	breaker.open.seconds	30	2025-09-04
Stream Processor	checkpoint.interval.seconds	60	2025-02-18
Stream Processor	allowed.lateness.minutes	2	2025-02-18
Stream Processor	deserialization.errors	route to DLQ	2026-09-09
Schema Registry	compatibility	BACKWARD_TRANSITIVE	2026-09-09
API Gateway	rate.limit.default.rps	6000 per client	2025-09-04
API Gateway	upstream.timeout.ms	300	2025-09-04
Case Management	first.touch.target.minutes	15 high value, 45 other	2026-05-05
Model Monitoring	psi.reference.window.days	28	2025-11-20

Key configuration parameters in production (as of 2026-09-28).

1.6 Kubernetes namespaces and resources

Namespace	Workloads	Requests per pod	Node group
edge	Kong Gateway	2 vCPU, 4 GiB	tier1-general
scoring	Orchestrator, FraudShield, Rules, Decision	Orchestrator 4 vCPU 8 GiB; FraudShield 2 vCPU 4 GiB	tier1-general and tier1-compute
features	Feature Service, Device Adapter, Merchant Risk	2 vCPU, 4 GiB	tier1-general
streaming	Flink JobManager and TaskManagers	TaskManager 8 vCPU 32 GiB	streaming
cases	Case Management, Label service	2 vCPU, 4 GiB	tier2-general
mlops	MLflow, monitoring service	2 vCPU, 8 GiB	tier2-general
observability	Prometheus, Grafana, Loki, Tempo	Varies	observability

Namespaces and resource requests.

2. Appendix B: Observability and testing

2.1 Metrics catalog

Metric	Type	Component	Used by
score_latency_seconds	Histogram	Orchestrator	PRFIP-LAT-001, PRFIP-LAT-002
risk_decisions_total{decision,degraded,decision_source}	Counter	Orchestrator	PRFIP-DEG-001, PRFIP-DEC-001
deadline_exceeded_total	Counter	Orchestrator	PRFIP-TMO-001, PRFIP-TMO-002
http_5xx_total	Counter	API Gateway	PRFIP-ERR-001
feature_lookup_seconds	Histogram	Feature Service	PRFIP-FS-LAT-001
redis_pool_in_use	Gauge	Feature Service	PRFIP-FS-POOL-001
redis_pool_limit	Gauge	Feature Service	PRFIP-FS-POOL-001
feature_lookups_per_request	Histogram	Scoring Service	Diagnostic (INC-2026-041)
feature_age_seconds	Histogram	Feature Service	PRFIP-FS-FRESH-001
inference_latency_seconds	Histogram	FraudShield	PRFIP-INF-001, PRFIP-INF-002
model_load_success	Gauge	FraudShield	PRFIP-FSVC-001
kafka_consumergroup_lag	Gauge	MSK exporter	PRFIP-KFK-001
dlq_messages_total	Counter	Stream Processor	PRFIP-KFK-002
flink_checkpoint_failures_total	Counter	Flink	PRFIP-FLK-001
flink_job_restarts_total	Counter	Flink	PRFIP-FLK-002
redis_memory_used_ratio	Gauge	ElastiCache	PRFIP-RDS-001
redis_replica_lag_seconds	Gauge	ElastiCache	PRFIP-RDS-002
score_psi	Gauge	Monitoring service	PRFIP-DRF-001, PRFIP-DRF-002
feature_psi{feature}	Gauge	Monitoring service	PRFIP-DRF-003
precision_at_decline_estimate	Gauge	Monitoring service	PRFIP-PRC-001
review_queue_open_cases	Gauge	Case Management	PRFIP-RVW-001
oldest_unassigned_case_minutes	Gauge	Case Management	PRFIP-RVW-001
case_api_5xx_total	Counter	Case Management	PRFIP-CASE-001
audit_pipeline_lag_seconds	Gauge	Audit service	PRFIP-AUD-001
dr_replication_lag_seconds	Gauge	Replication jobs	PRFIP-DR-001
certificate_days_to_expiry	Gauge	cert-exporter	PRFIP-CERT-001
vendor_breaker_state	Gauge	Device Adapter	Diagnostic
merchant_score_age_hours	Gauge	Merchant Risk	Alert at 36 hours

Key metrics.

2.2 Dashboards

Dashboard	Audience	Content
Scoring overview	SRE, Fraud Engineering	Traffic, latency percentiles, deadline exceeded, degraded share
Latency budget breakdown	SRE	Per-step latency for each downstream call
Feature Service	Feature Platform	Latency, pool utilization, lookups per request
Feature Store	Feature Platform	Redis CPU, memory, shards, failovers
Streaming	Streaming	Lag, checkpoints, restarts, DLQ
Feature freshness	SRE, Streaming	Feature age by family
Model health	Model owner	PSI, precision proxy, approval and review rates
Decisions by segment	Fraud Operations	Decision mix by channel, region, card type
Case queue	Fraud Operations	Open cases, age, throughput
Security	Security	Access to Confidential data, privileged sessions
DR readiness	SRE	Replication lag, standby capacity
Release	SRE	Canary metrics and comparison with baseline

Dashboards.

2.3 Testing strategy

Level	Scope	Tooling	Gate
Unit	Services and feature logic	JUnit, pytest	CI required
Contract	API Gateway, Kafka schemas (producer and consumer)	Pact, Schema Registry checks	CI required; contract tests between Payment Gateway and Flink added after INC-2026-057
Integration	Scoring path in staging	Replayed traffic	Before release
Load	Full scoring path at 120% of trailing 30-day peak with actual call pattern	k6, replay	Required for Tier 1 releases (RC-11)
Chaos	Fault injection in staging and quarterly in production	Chaos Mesh	Quarterly
Model	Validation, backtest, shadow	MLflow, in-house	Model Governance Committee
DR	Regional failover	Runbook RB-07	Quarterly
Security	Scans, penetration test	SAST, DAST, external tester	Annual and per release

Test levels.

2.4 Chaos experiments

ID	Experiment	Expected behavior	Last result
CH-01	Kill 50% of Feature Service pods	Level 2 for affected requests; recover within 2 minutes	Pass 2026-07-14
CH-02	Add 200 ms latency to Redis	Level 2 fallback; no deadline breaches above 0.5%	Pass 2026-07-14
CH-03	Exhaust Redis connection pool	Alert PRFIP-FS-POOL-001 fires within 5 minutes; fallback engaged	Pass 2026-09-02 (added after INC-2026-041)
CH-04	Stop FraudShield pods	Level 3 rules-only; alert Sev1	Pass 2026-07-14
CH-05	Break DeviceTrust vendor connectivity	Breaker opens; nulls; Level 1	Pass 2026-07-15
CH-06	Drop one Kafka AZ	No scoring impact; lag recovers	Pass 2026-07-15
CH-07	Publish incompatible schema in staging	Registry rejects registration	Pass 2026-09-10 (added after INC-2026-057)
CH-08	Poison record on transactions topic	Record goes to DLQ; job continues	Pass 2026-09-10
CH-09	Regional failover	See DR test history	Pass 2026-09-16
CH-10	Model artifact checksum mismatch	Pod refuses to start; alert	Pass 2026-07-16

Chaos experiments.

3. Appendix C: Model validation and evidence

3.1 Feature importance of FraudShield v4.2

Rank	Feature	Mean absolute SHAP share (%)
1	device_velocity_24h	7.9
2	merchant_risk_score	6.8
3	card_txn_count_1h	5.9
4	txn_amount_usd	5.4
5	amount_to_card_avg_ratio	4.7
6	device_trust_score	4.5
7	card_distinct_merchants_24h	4.1
8	ip_distinct_cards_24h	3.6
9	geo_velocity_kmh_card	3.4
10	device_distinct_cards_24h	3.3
11	card_amount_sum_24h	3.0
12	device_velocity_1h	2.9
13	email_first_seen_age_days	2.7
14	merchant_chargeback_rate_90d	2.6
15	is_3ds_authenticated	2.5
16	ip_reputation_score	2.4
17	card_first_seen_age_days	2.3
18	customer_txn_count_30d	2.2
19	card_bin_fraud_rate_90d	2.1
20	time_since_last_txn_card_sec	2.0
21	ip_txn_count_1h	1.9
22	device_first_seen_age_days	1.8
23	avs_result_code	1.7
24	email_reputation_score	1.6
25	customer_chargeback_count_180d	1.5
26	ip_geo_distance_to_billing_km	1.4
27	device_decline_ratio_7d	1.4
28	issuer_fraud_rate_90d	1.3
29	merchant_decline_ratio_24h	1.2
30	hour_of_day_local	1.1

Top 30 features by share of mean absolute SHAP value on the validation window (remaining 54 features share 29.3%).

3.2 Independent validation findings (VR-2026-007)

Model Risk & Validation, led by Aisha Khan, completed the independent validation of FraudShield v4.2 on 2026-03-06. All High findings were resolved before the Model Governance Committee met on 2026-03-18. Medium findings were accepted with monitoring, and Low findings were tracked.

ID	Finding	Severity	Resolution	Status
F-01	Calibration error higher for cross-border segment	Medium	Accepted; segment monitoring in place; improvement planned in v5.0	Accepted
F-02	Recent months rely on analyst labels with maturity bias	Medium	Maturity tagging; monthly re-estimation	Accepted
F-03	Possible leakage from customer_chargeback_count_180d	High	Window redefined to exclude post-event chargebacks; re-run	Closed
F-04	Handling of null device features under vendor outage	Low	Documented; tested in CH-05	Closed
F-05	Monotone constraints not documented in model card	Low	Model card updated	Closed
F-06	Threshold sensitivity around 0.85 decline boundary	Medium	Sensitivity analysis added; weekly monitoring	Accepted
F-07	Performance of reduced-feature model not documented	Medium	Section added; thresholds set stricter	Closed
F-08	Month-to-month stability	Low	Backtest added	Closed
F-09	Fairness proxy analysis by region incomplete	Medium	Analysis completed; no adverse difference above policy tolerance	Closed
F-10	Training run not fully reproducible	High	Seeds and data snapshot pinned; reproduced	Closed
F-11	Explanation to reason code mapping coverage 92%	Low	Mapping extended to 97%	Closed
F-12	Dependency on vendor device signals	Medium	Circuit breaker and null handling; vendor SLA review	Accepted

Findings of the independent validation of FraudShield v4.2.

3.3 Backtest by month

Month (2026)	Model in production	Recall at 1% FPR	Precision at decline	Fraud loss (bps of volume)	Review rate	Approval rate
January	v3.1	0.70	0.81	8.6	1.31%	96.4%
February	v3.1	0.71	0.82	8.4	1.29%	96.5%
March	v3.1	0.71	0.82	8.5	1.30%	96.4%
April	v4.2	0.77	0.85	8.0	1.22%	96.7%
May	v4.2	0.78	0.86	7.8	1.20%	96.8%
June	v4.2	0.78	0.86	7.7	1.19%	96.8%
July	v4.2	0.78	0.87	7.6	1.18%	96.9%
August	v4.2	0.76	0.85	7.9	1.27%	96.6%

Monthly performance. August includes the effect of INC-2026-041.

3.4 FraudShield v5.0 shadow evaluation (draft)

FraudShield v5.0 has scored a mirror of live traffic in shadow mode since 2026-08-19 under CHG-2026-0402. Its scores are logged only and never returned to callers. The results below are preliminary and do not constitute approval.

Week starting	Traffic scored (millions)	Recall at 1% FPR v5.0	Recall at 1% FPR v4.2	Inference p99 v5.0	Notes
2026-08-19	1,420	0.82	0.78	39 ms	First week; partial features
2026-08-26	2,310	0.83	0.78	41 ms	Full traffic mirrored
2026-09-02	2,290	0.81	0.77	44 ms	INC-2026-057 stale features affected both models
2026-09-09	2,330	0.84	0.78	41 ms	Stable
2026-09-16	2,350	0.83	0.78	42 ms	Stable
2026-09-23	2,320	0.83	0.78	41 ms	Cross-border ECE 0.031

Weekly shadow evaluation of FraudShield v5.0.

- Open concerns: inference latency p99 of 41 ms is more than double the 18 ms of v4.2 and needs optimization; cross-border calibration error of 0.031 is above the 0.020 target.
- Approval status: not approved. The proposal is on the agenda of MGC-2026-10 on 2026-10-15.
- Dependencies: Feature Service must serve the 28 new features and graph features require a daily graph job.

3.5 Feature membership by model version

FraudShield v3.1 used 61 features. The following features were added in v4.2 and were not used by v3.1: three_ds_version, browser_language_matches_country, email_local_part_entropy, order_has_digital_goods, shipping_method_expedited, time_since_last_txn_device_sec, card_distinct_countries_7d, amount_to_card_avg_ratio, device_distinct_cards_7d, device_decline_ratio_7d, ip_distinct_devices_24h, email_distinct_cards_7d, merchant_refund_ratio_30d, geo_velocity_kmh_card, device_trust_score, device_emulator_flag, device_rooted_flag, device_fingerprint_stability_7d, ip_asn_risk_tier, ip_is_datacenter, phone_reputation_score, customer_lifetime_chargeback_flag, card_first_seen_age_days.

Model	Features	Composition
v3.1 (archived)	61	All request, behavioral and enrichment features not marked v4.2 in the catalog
v4.2 (production)	84	61 from v3.1 plus 23 added in v4.2
v4.2-RF (fallback)	32	Request-only features
v5.0 (draft)	112	84 from v4.2 plus 28 new features

Feature counts by model.

4. Appendix D: Additional runbook procedures

4.1 RB-11: DeviceTrust vendor outage

Attribute	Value
Symptoms	Vendor circuit breaker open; reason code DEVICE_SIGNAL_MISSING share above 20%.
Impact	Device features are null (Level 1); accuracy slightly lower.

Steps

1. Confirm with the vendor status page and with vendor support that the outage is on the vendor side.
2. Verify that the circuit breaker is open and that no request errors occur.
3. Do not bypass the breaker or raise the vendor timeout; that would consume the request budget.
4. Notify Fraud Operations that device-based reason codes will be missing.
5. When the vendor recovers, the breaker closes after a successful probe; verify the missing share returns below 2%.

Escalation: Escalate to the Integration Lead (Naomi Fischer) for vendor contact.

4.2 RB-12: Threshold or rule change

Attribute	Value
Symptoms	A change to thresholds, threshold set or rule set is required.
Impact	Wrong thresholds change fraud losses and false declines immediately.

Steps

1. Threshold changes require approval of the Model Governance Committee (or its chair in an emergency, recorded afterwards); rule changes require the Fraud Operations Manager and Head of Fraud Engineering.
2. Raise a CHG with the expected effect estimated on the last 30 days of decisions.
3. Prepare the change as versioned configuration: a new threshold set or rule set version; never edit the running configuration.
4. Deploy to 5% of traffic first, monitoring approval, review and decline rates.
5. Promote to 100% after 4 hours if the effects match the estimate; otherwise revert to the previous version.
6. Record the change in the audit log and update the Confluence page for current status.

Escalation: Emergency rule changes for active attacks may be deployed by the Fraud Operations Manager with retrospective approval within one business day.

4.3 RB-13: Redis scaling and resharding

Attribute	Value
Symptoms	Memory alert PRFIP-RDS-001 or key cardinality growth.
Impact	Resharding briefly increases latency and can trigger fallbacks.

Steps

1. Raise a CHG; schedule in a low-traffic window, never during the Tuesday release window.
2. Check the current memory, connection and key distribution per shard.
3. Add shards using online resharding; migrate slots in batches of 100.
4. Watch Feature Service p99 and pool utilization during the migration; pause if p99 exceeds 20 ms.
5. After completion, check that all slots are balanced within 10%.
6. Update capacity documentation and the pool limit review.

Escalation: Stop the migration and page the Feature Platform Lead if degraded decisions exceed 0.5%.

4.4 RB-14: Kafka broker or partition problems

Attribute	Value
Symptoms	Under-replicated partitions, broker down, producer errors.
Impact	Async flows delayed; scoring path unaffected.

Steps

1. Check the MSK console and broker metrics; identify affected topics.
2. If one broker or AZ is down, verify that min.insync.replicas of 2 is still satisfied; producers continue.
3. Do not change partition counts of live topics without the streaming owner.
4. When the broker returns, wait for in-sync replicas before starting reassignments.
5. Check consumer lag and feature freshness after recovery.

Escalation: Escalate to Wei Zhang for partition reassignment.

4.5 RB-15: Feature backfill and replay

Attribute	Value
Symptoms	After a pipeline fault, features must be recomputed for a window.
Impact	Features are stale or wrong until backfilled.

Steps

1. Identify the start and end of the affected window from alerts and incident notes.
2. Choose the replay start offset in payments.transactions.v1 a full 24 hours before the window start so that 24-hour windows are correct.
3. Run the replay job with output to a separate topic; verify counts against the Silver layer.
4. Cut over the online store writer to the replay output; monitor freshness.
5. Record the backfill in the incident timeline and tell the model owner which decisions were affected.

Escalation: Backfills longer than 6 hours require the streaming owner and SRE Lead.

4.6 RB-16: Emergency access and security incident handoff

Attribute	Value
Symptoms	Suspected unauthorized access or a need for emergency production access.
Impact	Security and privacy exposure.

Steps

1. For suspected unauthorized access to Confidential data (PRFIP-SEC-001) declare SEV-1 and page the Principal Security Architect.
2. Preserve evidence: do not restart or delete resources before security agrees.
3. Revoke the suspected principal's sessions and rotate affected credentials.
4. For emergency access, use the break-glass role: it requires approval from the SRE Lead and is recorded for review within 24 hours.
5. Notify Data Governance (Rahul Mehta) if Confidential data may have been exposed.

Escalation: Legal and Privacy are engaged by the Principal Security Architect.

4.7 Shift handoff checklist

1. Open incidents and their severity, owner and next step.
2. Alerts silenced and the reason and expiry.
3. Changes scheduled for the next 24 hours, including canaries.
4. Current degraded decision rate and review queue size.
5. Any known drift or model concerns from the model owner.
6. Vendor or partner notifications.

4.8 Synthetic checks

Check	Frequency	Alerts on
Scoring request with a test transaction to /v1/risk/score	Every 30 seconds from 3 regions	Failure or latency above 200 ms
Feature Service lookup for a canary entity	Every 30 seconds	Failure or stale value
Kafka produce and consume round trip	Every minute	Round trip above 10 seconds
Case creation for a test REVIEW decision	Every 5 minutes	Case not created within 30 seconds
DR region readiness	Hourly	Standby unhealthy
Certificate expiry	Daily	Expiry under 14 days

Synthetic monitoring.

5. Appendix E: Threat model and security assessments

5.1 Threat model

ID	Threat	Category	Asset	Controls	Residual risk
T-01	Stolen partner credentials used to call the scoring API	Spoofing	Scoring API	mTLS, OAuth client credentials, rate limits, IP allowlist	Low
T-02	Attacker probes the API to learn thresholds	Information disclosure	Model behavior	Rate limits, no score returned to merchants, reason codes internal only	Medium
T-03	Tampering with features in the online store	Tampering	Redis	Network isolation, ACLs, no human access	Low
T-04	Poisoning of training labels	Tampering	Training data	Label service controls, sampling review, segregation of duties	Medium
T-05	Malicious model artifact substituted	Tampering	Model Registry	Checksum verification, registry lifecycle controls	Low
T-06	Insider export of Confidential data	Information disclosure	Snapshots	Secure views, logging, alert PRFIP-SEC-001, no export without approval	Medium
T-07	Denial of service on scoring	Denial of service	Availability	Rate limits, autoscaling, degradation ladder	Medium
T-08	Vendor compromise	Tampering	DeviceTrust signals	Signal validation, breaker, model tolerant to nulls	Medium
T-09	Privilege escalation through CI/CD	Elevation of privilege	Pipelines	Signed builds, protected branches, JIT access	Low
T-10	Replay of scoring requests	Spoofing	Decisions	Idempotency keys, request timestamps	Low
T-11	Card testing to enumerate valid cards	Fraud	Payment flow	Rules R-004 and R-007, BIN attack score in v5.0 draft	Medium
T-12	Log leakage of identifiers	Information disclosure	Logs	Field masking, log review	Low
T-13	Schema change breaks consumers (accidental)	Availability	Feature pipeline	Compatibility enforcement (ADR-012), DLQ	Low
T-14	Loss of audit evidence	Repudiation	Audit store	Object Lock, Azure immutable copy	Low

Threat model summary.

5.2 Penetration test summary (May 2026)

ID	Finding	Severity	Status
PT-01	Verbose error messages on a partner endpoint	Low	Closed 2026-06-04
PT-02	Missing rate limit on health endpoint	Low	Closed 2026-06-02
PT-03	Overly broad IAM role on a build runner	Medium	Closed 2026-06-10
PT-04	Grafana viewer could list dashboards with entity identifiers	Medium	Closed 2026-06-12
PT-05	Outdated base image in a non-production namespace	Low	Closed 2026-06-20
PT-06	No finding above Medium in the scoring path	Info	n/a

Findings of the May 2026 penetration test.

5.3 Third-party assessment: DeviceTrust

Area	Assessment (2026-04)	Outcome
Certification	SOC 2 Type II current	Satisfactory
Data shared	Device fingerprint hash and IP address only	Minimal
Availability	99.9% SLA; 99.94% measured over 12 months	Satisfactory
Subprocessors	Two, listed and approved	Satisfactory
Breach notification	Within 48 hours	Satisfactory
Exit plan	Documented; model tolerant of null device features	Satisfactory

Vendor assessment of DeviceTrust.

6. Appendix F: Earlier 2026 incident summaries

6.1 INC-2026-027: Case creation delay (2026-05-05, SEV-2)

Case creation was delayed by up to 40 minutes after a Kafka consumer group rebalance in Case Management. The consumer used a session timeout shorter than its longest processing step, causing repeated rebalances. Review decisions continued to be made and were queued. The consumer session timeout was raised from 10 to 45 seconds, processing was made asynchronous, and alert PRFIP-CASE-001 was added. Owner of actions: Tom Alvarez and the Case Management team. Status: closed.

6.2 INC-2026-033: Elevated timeouts after certificate rotation (2026-06-16, SEV-3)

After the scheduled mTLS certificate rotation, one Orchestrator pool retained an expired intermediate certificate, causing handshake failures to the Feature Service for about 6% of pods. Deadline exceeded rate reached 0.9% for 22 minutes. The pods were restarted and the rotation procedure was extended with a verification step and the alert PRFIP-CERT-001 (14 days). Status: closed.

6.3 INC-2026-019: DeviceTrust vendor timeouts (2026-03-27, SEV-3)

The vendor experienced a regional outage lasting 47 minutes. The circuit breaker opened after five consecutive timeouts, device features were null and the reason code DEVICE_SIGNAL_MISSING was recorded for 31% of decisions. There was no measurable change in approval rate and a small increase in review rate. This incident confirmed the design of the breaker and led to chaos experiment CH-05. Status: closed.

7. Appendix G: Data Platform Architecture

Attribute	Value
Document	Data Platform Architecture
Version / status	v1.8 / CURRENT (Approved)
Effective date	2026-08-05
Owner	Priya Raman (platform) and Rahul Mehta (governance)
Supersedes	v1.7 (2026-03-02, ARCHIVED)

7.1 Data zones

The data platform stores the history that the fraud models and analytics depend on. It is organized in four zones on Amazon S3 and Snowflake. Each zone has a defined purpose, quality level and access rule. Data moves in one direction, from raw to curated, and every transformation is reproducible from the raw zone.

Zone	Purpose	Format	Quality	Access	Retention
Raw (Bronze)	Immutable copy of events as received	Parquet on S3, partitioned by event date and topic	As received; malformed records quarantined	Platform engineers and service accounts	13 months hot, 7 years archive
Cleansed (Silver)	Validated, deduplicated, tokenized	Iceberg tables	Schema-validated; deduplicated by transaction_id	Data scientists via secure views (Confidential role)	25 months
Curated (Gold)	Feature tables, labeled datasets, reporting marts	Iceberg and Snowflake tables	Data quality rules enforced; failures block publish	Analysts, data scientists, model pipelines	25 months
Sandbox	Exploration	Snowflake schemas	None	Individual users; Internal data only unless approved	90 days

Data zones.

7.2 Table catalog

Table	Zone	Grain	Source	Refresh	Classification
bronze.transactions	Bronze	One row per transaction event	payments.transactions.v1	Streaming, under 5 minutes	Confidential
bronze.scores	Bronze	One row per score	risk.scores.v1	Streaming	Confidential
bronze.decisions	Bronze	One row per decision	risk.decisions.v1	Streaming	Confidential
bronze.feature_snapshots	Bronze	One row per request feature vector	features.snapshots.v1	Streaming, under 15 minutes	Confidential
bronze.device_signals	Bronze	One row per vendor response	devices.signals.v1	Streaming	Confidential
bronze.chargebacks	Bronze	One row per chargeback record	Chargeback files	Daily	Confidential
bronze.case_events	Bronze	One row per case event	fraud.cases.v1	Streaming	Confidential
silver.transactions	Silver	One row per unique transaction	bronze.transactions	Hourly	Confidential
silver.decisions	Silver	One row per decision with model and threshold version	bronze.decisions	Hourly	Confidential
silver.entities_card	Silver	One row per card token	silver.transactions	Daily	Confidential
silver.entities_device	Silver	One row per device	silver.transactions	Daily	Confidential
silver.labels	Silver	One row per labeled transaction with maturity	Chargebacks and cases	Daily	Confidential
gold.features_offline	Gold	Point-in-time feature values per entity and time	Silver, snapshots	Daily	Confidential
gold.training_v42	Gold	Training set for FraudShield v4.2	gold.features_offline, silver.labels	Frozen	Confidential
gold.training_v50_draft	Gold	Working training set for FraudShield v5.0	gold.features_offline, silver.labels	Weekly	Confidential
gold.merchant_risk	Gold	One row per merchant and day	silver.transactions, silver.labels	Daily	Confidential
gold.model_monitoring	Gold	Drift and performance by hour	Scores, snapshots, labels	Hourly	Confidential
gold.ops_dashboard	Gold	Aggregated operational KPIs	Decisions, cases	Hourly	Internal
gold.case_kpis	Gold	Case throughput and resolution times	Case events	Hourly	Internal
gold.finance_fraud_loss	Gold	Fraud loss by day and merchant segment	Labels, chargebacks	Daily	Confidential

Principal tables.

7.3 Orchestration

DAG	Schedule	SLA	Owner	Purpose
bronze_to_silver_transactions	Hourly	20 minutes after the hour	Data Engineering	Validate and deduplicate transactions
silver_entities_daily	Daily 02:00 UTC	06:00 UTC	Data Engineering	Entity tables
gold_features_offline_daily	Daily 03:00 UTC	07:00 UTC	Feature Platform	Point-in-time feature tables
merchant_risk_daily	Daily 01:30 UTC	04:00 UTC	Sarah Johnson	Compute merchant_risk_score
merchant_risk_hourly	Hourly	15 minutes	Sarah Johnson	Incremental merchant score refresh
labels_daily	Daily 05:00 UTC	08:00 UTC	Sarah Johnson	Chargeback and analyst label consolidation
graph_features_daily_draft	Daily 04:00 UTC	09:00 UTC	Sarah Johnson	Graph features for v5.0 draft
monitoring_hourly	Hourly	10 minutes	Sarah Johnson	PSI and precision proxy
retention_enforcement_daily	Daily 23:00 UTC	24 hours	Rahul Mehta	Apply retention and erasure
access_review_quarterly	Quarterly	Two weeks	Rahul Mehta	Access recertification extract

Orchestration DAGs.

7.4 Data quality rules

ID	Rule	Applies to	Threshold	On failure
DQ-01	transaction_id unique	silver.transactions	0 duplicates	Block publish
DQ-02	amount_minor_units is an integer greater than 0	silver.transactions	100%	Quarantine record
DQ-03	currency_code valid ISO 4217	silver.transactions	100%	Quarantine record
DQ-04	event_time not in the future	silver.transactions	99.99%	Alert
DQ-05	card_token_id not null	silver.transactions	100%	Quarantine record
DQ-06	No value matching PAN pattern in any column	All bronze and silver	0	Sev1 security alert
DQ-07	Feature null rate within 3 sigma of 28-day mean	gold.features_offline	Per feature	Block publish
DQ-08	Row count within 20% of same weekday last week	bronze.transactions	Daily	Alert
DQ-09	Label maturity tag present	silver.labels	100%	Block publish
DQ-10	Every decision has model_version and threshold_set_version	silver.decisions	100%	Block publish
DQ-11	Snapshot to decision join rate	bronze.feature_snapshots	above 99.9%	Alert
DQ-12	merchant_risk_score in range 0 to 1	gold.merchant_risk	100%	Block publish
DQ-13	Training-serving skew per feature	Snapshots vs offline	PSI below 0.10	Alert to Feature Platform
DQ-14	Schema matches registered schema version	All bronze	100%	Quarantine and DLQ

Data quality rules.

7.5 Lineage and ownership

Lineage is recorded in the data catalog from source topic to consuming model or report. For every model version the catalog records the training table and its snapshot date, so that any decision can be traced back to the data that shaped the model that made it. Ownership of each table is recorded and reviewed quarterly by Data Governance.

8. Appendix H: Production Support Guide

Attribute	Value
Document	Fraud Platform Production Support Guide
Version / status	v2.4 / CURRENT (Approved)
Effective date	2026-09-10
Owner	Daniel Okafor

This guide answers the questions that support engineers and fraud analysts ask most often. It complements the Production Runbook: use the runbook for incidents and this guide for everyday questions.

Question	Answer
What is the current production fraud model?	FraudShield v4.2, effective 2026-04-06, owner Sarah Johnson. Version v3.1 is archived; v5.0 is a draft in shadow evaluation.
Which thresholds apply to decisions?	Threshold set TS-4.2.1: DECLINE at or above 0.85, REVIEW from 0.60 to below 0.85, STEP_UP from 0.50 to below 0.60 for e-commerce without 3DS, otherwise APPROVE.
What is the timeout of /v1/risk/score?	The server-side deadline is 250 ms. After it the API returns HTTP 200 with a rules-only decision and degraded=true. The Payment Gateway client timeout is 400 ms.
What does degraded=true mean?	The decision was made under a degradation mode: the reduced-feature model v4.2-RF or rules only. Check decision_source and the reason codes.
Why is risk_score null?	The decision was rules-only (decision_source RULES_ONLY), because FraudShield was unavailable or the 250 ms deadline was reached.
What happens when the Feature Service is unavailable?	The Orchestrator uses FraudShield v4.2-RF with the 32 request-only features and stricter thresholds. Procedure RB-02 applies.
What should I do if FraudShield stops responding?	Follow RB-01: declare SEV-1, check pods and the latest release, roll back the Scoring Service if a release preceded the fault, never roll back to v3.1.
Can I roll back to FraudShield v3.1?	No. v3.1 is archived and was removed from the serving registry on 2026-07-06.
What are the conditions for a model rollback?	Score PSI above 0.25 for 2 hours with lower precision; precision at decline below 65% for 24 hours; inference p99 above 60 ms for 15 minutes; approval rate deviation above 3 points for 30 minutes.
Who approves a model rollback?	The Incident Commander together with the model owner (Sarah Johnson) or delegate.
What data classification applies to transaction-level features?	Confidential. Static reference features such as MCC and BIN country are Internal.
Does PRFIP store card numbers?	No. It receives only a token (card_token_id), the BIN and last four digits.
Where does merchant_risk_score originate?	The Merchant Risk Service computes it in a daily batch job with hourly incremental refresh; it is stored in the online store.
What does device_velocity_24h represent?	The number of prior transactions from the same device in the last 24 hours, computed by the stream processor.
How fresh are streaming features?	p95 under 60 seconds. An alert pages the on-call engineer when p95 age exceeds 120 seconds.
How long do streaming features stay in the online store if the pipeline stops?	6 hours (TTL); batch features 36 hours.
What was the cause of the August 2026 latency incident?	INC-2026-041: Scoring Service SS-2.14.0 raised Feature Service lookups per request from 6 to 14 without raising the Redis connection pool (200 per pod). The pool was exhausted at peak. Fixed by SS-2.14.1 (pipelined lookups, pool 480).
What was the cause of INC-2026-057?	A breaking schema change (amount from integer to decimal string) on payments.transactions.v1 registered without compatibility enforcement; Flink failed and features became stale for 3 hours 12 minutes.
Why did the August incident not cause errors for customers?	The degradation ladder returned reduced-feature or rules-only decisions with HTTP 200.
What are the RTO and RPO for the scoring path?	RTO 30 minutes and RPO 5 minutes, with a warm standby in us-west-2.
Is active-active multi-region approved?	No. ADR-010 is PROPOSED and was deferred at ARB-2026-06.
When are Tier 1 releases deployed?	Tuesdays 10:00 to 12:00 UTC, with a canary of at least 24 hours that includes a peak.
What load test is required before a Tier 1 release?	At 120% of the trailing 30-day peak with the release's actual call pattern (checklist item RC-11).

Question	Answer
Who owns the Feature Service?	Priya Raman, Feature Platform Lead.
Who owns the Production Runbook?	Daniel Okafor, SRE Lead.
Who owns the security architecture?	Elena Petrova, Principal Security Architect.
How long are audit logs kept?	7 years, immutable.
How many features does FraudShield v4.2 use?	84: 32 request features, 33 streaming behavioral features and 19 enrichment features.
Which documents are authoritative when versions differ?	The document with status CURRENT and the latest effective date. DRAFT and PROPOSED documents are not production truth; ARCHIVED documents describe past behavior.
What should I do if the manual review queue is growing?	See RB-09 and check whether degraded decisions are inflating the queue; ask the Fraud Operations Manager to add analysts.
Where are new rules proposed?	Fraud Operations proposes; the Fraud Operations Manager and Head of Fraud Engineering approve; rules have an owner and expiry date.

Frequently asked questions.

8.1 Contacts

Area	Contact	Channel
Scoring platform	Marcus Lee	#prfip-platform
Feature Service and store	Priya Raman	#prfip-features
Model and thresholds	Sarah Johnson	#prfip-models
Streaming	Wei Zhang	#prfip-streaming
SRE and incidents	Daniel Okafor	#prfip-oncall and PagerDuty
Security	Elena Petrova	#prfip-security
Fraud Operations	Tom Alvarez	#fraud-ops
Data governance	Rahul Mehta	#data-governance
Payment Gateway integration	Naomi Fischer	#pgw-integration
Model validation	Aisha Khan	#model-risk

Contacts.

8.2 Release history reference

Release	Date	Description	Change	Approval
REL-2026-014	2026-04-06	FraudShield v4.2 production release	CHG-2026-0211	Approved MGC-2026-03; RRR-2026-014
REL-2026-021	2026-05-19	Scoring Service SS-2.12.0: reason code improvements	CHG-2026-0290	Approved
REL-2026-027	2026-06-23	Feature Service FS-1.9.0: added device velocity windows	CHG-2026-0334	Approved
REL-2026-031	2026-08-04	Scoring Service SS-2.14.0: 14 per-family lookups	CHG-2026-0388	Approved with conditions; rolled back 2026-08-11 (INC-2026-041)
REL-2026-032	2026-08-11	Rollback to Scoring Service SS-2.13.2	Emergency change	Approved by incident commander
REL-2026-033	2026-08-25	Scoring Service SS-2.14.1: pipelined lookups, pool 480	CHG-2026-0411	Approved; RRR-2026-033
REL-2026-034	2026-08-19	FraudShield v5.0 shadow deployment (no customer impact)	CHG-2026-0402	Approved
REL-2026-036	2026-09-09	Schema Registry BACKWARD_TRANSITIVE enforcement; Flink DLQ	CHG-2026-0421	Approved

Recent releases.